Clinical interpretation — sub-M2030 ses-1454

Chronic (>6 mo) 3839

days post-onset

Lesion mask: **Manual** · recovery stage: **Chronic (>6 mo)** · research use only, not a diagnosis
Selected evidence scope: **approved, in_review, draft** · (all confidence levels included) · atlas (lookup): AAL, HarvardOxford, Destrieux, Schaefer400, JHU, XTRACT · atlas (display): HarvardOxford

QC status: No QC checkpoint choice is recorded.

Research use only — not a diagnosis. This report combines automated lesion segmentation with literature associations from the aphasia knowledge base. It does not diagnose aphasia, predict an individual's outcome or prescribe therapy; any clinical use requires review by a qualified clinician.

- **Lesion mask:** Manual (MNI) `ExpertMask_in_MNI.nii.gz`. QC: not rated in the QC checkpoint.
- **How this lesion mask was made:**
  1. Expert-drawn mask sub-M2030_ses-1455_acq-spc3_run-3_T2w_desc-lesion_mask.nii.gz, drawn on sub-M2030_ses-1455_acq-spc3_run-3_T2w.nii.gz; that scan was rigidly registered to the T1w sub-M2030_ses-1454_acq-tfl3p2_run-3_T1w.nii.gz (SimpleITK, Mattes mutual information, seed 1729) and the mask resampled with nearest neighbour.
  2. Warped to the FSL MNI152 template (MNI152NLin6Asym, 1 mm) with this T1w's ANTs registration (ants/2.6.5: affine, then SyN, with the SynthStroke lesion excluded from the cost function, seed 1729), nearest-neighbour resampling.
- **Template space:** Lesion masks are on the FSL MNI152 template (MNI152NLin6Asym, 1 mm); the registration used for this mask is named above. Atlases are resampled to the lesion by world coordinates. Overlaps still depend on how well this individual brain was registered to MNI152, which matters most for small or thin regions.
- **Knowledge base:** version b15b88b (2026-10-22) with uncommitted changes · 116 papers · 507 extracted findings (47 approved by a named reviewer) in 272 entries (171 regions, 18 impairments, 21 therapies, 62 predictors) · statuses used: approved, in_review, draft
- **Generated:** 2026-10-21 22:13 · KB scope: approved, in_review, draft (all confidence levels included)

Could not render lesion overlay: [Errno 2] No such file or directory: '/home/jovyan/neurodesktop-storage/calmar/reports/jobs/59f156908e634f39bfa7b683d13699f7/result/lesion_overlay.png'

Damaged regions (HarvardOxford, ≥5.0%)

This table locates the lesion in the selected display atlas. It is an anatomical overview; the evidence interpretation appears in the sections below.

HarvardOxford region	Lesion side	% of lesion in ROI	% of ROI lesioned	
Temporal Pole	Left	7.4%	38.9%	1,833 vox
Insular Cortex	Left	4.5%	47.5%	1,113 vox
Central Opercular Cortex	Left	4.0%	52.2%	984 vox
Supramarginal Gyrus, anterior division	Left	3.7%	52.6%	916 vox
Superior Temporal Gyrus, posterior division	Left	3.5%	47.9%	877 vox
Supramarginal Gyrus, posterior division	Left	3.4%	37.3%	843 vox
Middle Temporal Gyrus, posterior division	Left	3.3%	32.4%	819 vox
Angular Gyrus	Left	2.9%	30.3%	731 vox
Parietal Opercular Cortex	Left	2.3%	51.5%	576 vox
Planum Temporale	Left	2.1%	56.0%	533 vox
Middle Temporal Gyrus, anterior division	Left	1.8%	52.1%	442 vox
Inferior Frontal Gyrus, pars opercularis	Left	1.8%	31.5%	436 vox
Frontal Opercular Cortex	Left	1.5%	52.7%	362 vox
Planum Polare	Left	1.4%	46.1%	345 vox
Heschl's Gyrus (includes H1 and H2)	Left	1.3%	53.0%	320 vox

Lesion side is measured from the actual mask overlap; bilateral entries show left/right counts.

Behavioural / clinical data

These participant measures provide context for the evidence lookup. Missing values remain missing and are not inferred.

WAB-AQ	51.5
WAB type	Broca `[flu − · comp + · rep −]`
Age at stroke	54
Sex	M
Days to WAB	3839
Lesion volume	199.2 mL

Mapped to KB predictors: severity_metric, age, time_post_onset, sex, lesion_volume

Lesion location: what the evidence says

This section connects the participant's lesion location to published group-level findings. Start with the outcome summaries, then open a damaged region to inspect each supporting, null or mixed finding.

A region counts as damaged when it holds ≥5% of the lesion or ≥25% of it is destroyed. Regions are matched to the knowledge base through the curated atlas mapping (`aphasia-kb/atlas_mapping/`). Of the knowledge base's 348 location findings, **252** concern regions this lesion damages, 68 concern regions it does not damage, and 18 concern regions no atlas can represent (e.g. the extreme capsule).

▶ **What does the display signal mean?**

Display signal: a project-specific ordering heuristic, not an evidence grade. It combines lesion overlap with stored finding strength, study evidence class, extraction confidence and relationship direction. Counts do not measure certainty, and records may share studies or cohorts.

IMPAIRMENT

Semantics (semantic processing)

Display signal: very strong / replicated

Across 36 distinct matched findings, 34 associations and 2 null findings. These are group associations, not a participant-level prediction.

ASSOCIATIONS	NULL FINDINGS
34	**2**

Region preview: Left Angular Gyrus, Left Anterior Temporal Lobe, Left IFG Pars Triangularis (BA 45)

▶ **Show all 14 implicated regions**

▶ **Sources (17 citation keys)**

All implicated regions: Left Angular Gyrus, Left Anterior Temporal Lobe, Left IFG Pars Triangularis (BA 45), Left Inferior Fronto-Occipital Fasciculus, Left Inferior Longitudinal Fasciculus, Left Middle Longitudinal Fasciculus, Left Middle Temporal Gyrus (Ventral Stream Node), Left Parietal Association Cortex, Left Superior Temporal Gyrus (Dorsal Stream Node), Left Superior Temporal Gyrus (Wernicke's Area and Planum Temporale), Left Superior Temporal Sulcus, Left Uncinate Fasciculus, Left anterior middle temporal gyrus, Left temporal pole

Sources: @Alyahya2018NounVerb, @Alyahya2020, @Baldo2013, @Bates2003, @Binder2009, @Breier2008, @Butler2014, @Catani2008, @DeWitt2012, @Dick2012, @Dronkers2004, @Fridriksson2013, @Halai2017, @Han2013, @Hillis2001, @Ivanova2016, @Schwartz2009

IMPAIRMENT

Phonological production

Display signal: very strong / replicated

Across 19 distinct matched findings, 18 associations and 1 null findings. These are group associations, not a participant-level prediction.

ASSOCIATIONS	NULL FINDINGS
18	**1**

Region preview: Left Angular Gyrus, Left Arcuate Fasciculus and Superior Longitudinal Fascicle (Dorsal Language Pathway), Left IFG Pars Opercularis (Broca's area, dorsal stream hub)

▶ **Show all 12 implicated regions**

▶ **Sources (10 citation keys)**

All implicated regions: Left Angular Gyrus, Left Arcuate Fasciculus and Superior Longitudinal Fascicle (Dorsal Language Pathway), Left IFG Pars Opercularis (Broca's area, dorsal stream hub), Left Inferior Longitudinal Fasciculus, Left Posterior Superior Temporal Gyrus, Left Superior Longitudinal Fasciculus, Left Superior Temporal Gyrus (Dorsal Stream Node), Left Supramarginal Gyrus, Left Supramarginal Gyrus and Premotor Cortex (supra-Sylvian), Left arcuate fasciculus, Left parietal operculum cortex, Left posterior supramarginal gyrus

Sources: @Alyahya2018NounVerb, @Alyahya2020, @Breier2008, @Butler2014, @Dick2012, @Halai2017, @Henseler2014, @Ivanova2016, @Kummerer2013, @Schwartz2012

IMPAIRMENT

Broca's aphasia

Display signal: very strong / replicated

Across 11 distinct matched findings, 9 associations, 1 null findings and 1 mixed results. These are group associations, not a participant-level prediction.

ASSOCIATIONS	NULL FINDINGS	MIXED RESULTS
9	**1**	**1**

Region preview: Left Heschl's Gyrus, Left IFG Pars Opercularis (Broca's area, dorsal stream hub), Left Superior Temporal Gyrus (Dorsal Stream Node)

▶ **Show all 7 implicated regions**

▶ **Sources (3 citation keys)**

All implicated regions: Left Heschl's Gyrus, Left IFG Pars Opercularis (Broca's area, dorsal stream hub), Left Superior Temporal Gyrus (Dorsal Stream Node), Left Supramarginal Gyrus, Left anterior arcuate fasciculus, Left arcuate fasciculus, Left rolandic operculum

Sources: @Henseler2014, @Ochfeld2010, @Yourganov2015Predicting

IMPAIRMENT

Fluency (speech fluency)

Display signal: very strong / replicated

Across 20 distinct matched findings, 17 associations and 3 null findings. These are group associations, not a participant-level prediction.

ASSOCIATIONS	NULL FINDINGS
17	**3**

Region preview: Left Arcuate Fasciculus and Superior Longitudinal Fascicle (Dorsal Language Pathway), Left IFG Pars Opercularis (Broca's area, dorsal stream hub), Left IFG Pars Triangularis (BA 45)

▶ **Show all 11 implicated regions**

▶ **Sources (8 citation keys)**

All implicated regions: Left Arcuate Fasciculus and Superior Longitudinal Fascicle (Dorsal Language Pathway), Left IFG Pars Opercularis (Broca's area, dorsal stream hub), Left IFG Pars Triangularis (BA 45), Left Inferior Fronto-Occipital Fasciculus, Left Posterior Superior Temporal Gyrus, Left Superior Temporal Gyrus (Wernicke's Area and Planum Temporale), Left Uncinate Fasciculus, Left anterior arcuate fasciculus, Left arcuate fasciculus, Left precentral gyrus of the insula, Left rolandic operculum

Sources: @Alyahya2018NounVerb, @Bates2003, @Bonilha2014, @Fridriksson2013, @GajardoVidal2021, @Hope2013, @Li2017, @Yourganov2016

IMPAIRMENT

Form-to-meaning processing (ventral stream)

Display signal: very strong / replicated

Across 9 distinct matched findings, 9 associations. These are group associations, not a participant-level prediction.

ASSOCIATIONS
9

Region preview: Left Angular Gyrus, Left Posterior Superior Temporal Gyrus, Left Superior Temporal Gyrus (Dorsal Stream Node)

▶ **Show all 8 implicated regions**

▶ **Sources (4 citation keys)**

All implicated regions: Left Angular Gyrus, Left Posterior Superior Temporal Gyrus, Left Superior Temporal Gyrus (Dorsal Stream Node), Left Uncinate Fasciculus, Left anterior middle temporal gyrus, Left arcuate fasciculus, Left posterior middle temporal gyrus, Left posterior supramarginal gyrus

Sources: @Breier2008, @Fridriksson2013, @Fridriksson2018, @Henseler2014

IMPAIRMENT

Phonological recognition

Display signal: very strong / replicated

Across 6 distinct matched findings, 6 associations. These are group associations, not a participant-level prediction.

ASSOCIATIONS
6

Region preview: Left Heschl's Gyrus, Left Posterior Planum Temporale (Area Spt), Left Posterior Superior Temporal Gyrus

▶ **Show all 4 implicated regions**

▶ **Sources (4 citation keys)**

All implicated regions: Left Heschl's Gyrus, Left Posterior Planum Temporale (Area Spt), Left Posterior Superior Temporal Gyrus, Left Superior Temporal Gyrus (Dorsal Stream Node)

Sources: @Alyahya2018NounVerb, @Alyahya2020, @DeWitt2012, @Thye2018

IMPAIRMENT

Global aphasia

Display signal: very strong / replicated

Across 6 distinct matched findings, 6 associations. These are group associations, not a participant-level prediction.

ASSOCIATIONS
6

Region preview: Left IFG Pars Triangularis (BA 45), Left Uncinate Fasciculus, Left anterior arcuate fasciculus

▶ **Show all 6 implicated regions**

▶ **Sources (1 citation key)**

All implicated regions: Left IFG Pars Triangularis (BA 45), Left Uncinate Fasciculus, Left anterior arcuate fasciculus, Left precentral gyrus of the insula, Left putamen, Left temporal pole MTG

Sources: @Yourganov2015Predicting

IMPAIRMENT

Syntactic Processing Deficits in Aphasia

Display signal: very strong / replicated

Across 11 distinct matched findings, 7 associations, 3 null findings and 1 mixed results. These are group associations, not a participant-level prediction.

ASSOCIATIONS	NULL FINDINGS	MIXED RESULTS
7	**3**	**1**

Region preview: Left Angular Gyrus, Left Anterior Temporal Lobe, Left IFG Pars Opercularis (Broca's area, dorsal stream hub)

▶ **Show all 7 implicated regions**

▶ **Sources (4 citation keys)**

All implicated regions: Left Angular Gyrus, Left Anterior Temporal Lobe, Left IFG Pars Opercularis (Broca's area, dorsal stream hub), Left IFG Pars Triangularis (BA 45), Left Posterior Superior Temporal Gyrus, Left Superior Temporal Gyrus (Dorsal Stream Node), Left Superior Temporal Gyrus (Wernicke's Area and Planum Temporale)

Sources: @Dronkers2004, @Matchin2020, @Rogalsky2018, @Wilson2023

IMPAIRMENT

Speech repetition

Display signal: very strong / replicated

Across 7 distinct matched findings, 7 associations. These are group associations, not a participant-level prediction.

ASSOCIATIONS
7

Region preview: Left Arcuate Fasciculus and Superior Longitudinal Fascicle (Dorsal Language Pathway), Left IFG Pars Opercularis (Broca's area, dorsal stream hub), Left Posterior Superior Temporal Gyrus

▶ **Show all 7 implicated regions**

▶ **Sources (3 citation keys)**

All implicated regions: Left Arcuate Fasciculus and Superior Longitudinal Fascicle (Dorsal Language Pathway), Left IFG Pars Opercularis (Broca's area, dorsal stream hub), Left Posterior Superior Temporal Gyrus, Left Superior Temporal Gyrus (Dorsal Stream Node), Left Supramarginal Gyrus, Left Temporoparietal Junction, Left arcuate fasciculus

Sources: @Fridriksson2010, @Saur2008, @Yourganov2016

IMPAIRMENT

Auditory comprehension

Display signal: very strong / replicated

Across 6 distinct matched findings, 6 associations. These are group associations, not a participant-level prediction.

ASSOCIATIONS
6

Region preview: Left Middle Temporal Gyrus (Ventral Stream Node), Left Posterior Superior Temporal Gyrus, Left Superior Temporal Gyrus (Wernicke's Area and Planum Temporale)

▶ **Show all 4 implicated regions**

▶ **Sources (5 citation keys)**

All implicated regions: Left Middle Temporal Gyrus (Ventral Stream Node), Left Posterior Superior Temporal Gyrus, Left Superior Temporal Gyrus (Wernicke's Area and Planum Temporale), Left Superior Temporal Sulcus

Sources: @Bonilha2014, @DeWitt2012, @Robson2019, @Saur2008, @Yourganov2016

Evidence by damaged region

Open a region to inspect distinct findings, grouped by outcome. A limited citation base means 2 citation keys or fewer; it is not a certainty rating.

Left Posterior Superior Temporal Gyrus (left)

3.5% of the lesion · 100% of the region destroyed · 17 distinct findings

Also recorded in the knowledge base as: Left posterior superior temporal gyrus

Sentence comprehension 1 finding

Damage to left posterior STG is the strongest independent predictor of sentence comprehension impairment in chronic left-hemisphere stroke aphasia, accounting for 60% of variance in the multivariate RLSM analysis.

@Fridriksson2018Anatomy · Study method: LSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Strong**	**High**

Entry status: approved · Review provenance: legacy approval · Reviewer: claude-cowork

▶ **Full citation for @Fridriksson2018Anatomy**

▶ **Study context, limitations, review and source passages**

Auditory word recognition 1 finding

Damage to left STG and posterior STG is the strongest predictor of impaired auditory word recognition in chronic left-hemisphere stroke aphasia, consistent with the ventral stream's role in auditory-to-meaning mapping.

@Fridriksson2018Anatomy · Study method: LSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Strong**	**High**

Entry status: approved · Review provenance: legacy approval · Reviewer: claude-cowork

▶ **Full citation for @Fridriksson2018Anatomy**

▶ **Study context, limitations, review and source passages**

Naming 2 findings

Naming impairment (PNT correct) in chronic left-hemisphere stroke aphasia is associated with damage to an extensive cortical network including both ventral stream (posterior STG) and dorsal stream (precentral gyrus) regions, with no single dominant lesion location.

@Fridriksson2018Anatomy · Study method: LSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: approved · Review provenance: legacy approval · Reviewer: claude-cowork

▶ **Full citation for @Fridriksson2018Anatomy**

▶ **Study context, limitations, review and source passages**

Damage to the left posterior superior temporal gyrus is a significant predictor of naming in chronic aphasia.

@Yourganov2016 · Study method: MLPA · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Yourganov2016**

▶ **Study context, limitations, review and source passages**

Auditory comprehension 2 findings

Structural disconnection of left BA 22 (posterior superior temporal gyrus / Wernicke's area) is independently associated with impaired auditory comprehension of sequential commands; BA 22 cortical necrosis alone is not an independent predictor of comprehension.

@Bonilha2014 · Study method: disconnectome · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**Medium**

Entry status: approved · Review provenance: legacy approval · Reviewer: claude-cowork

▶ **Full citation for @Bonilha2014**

▶ **Study context, limitations, review and source passages**

Damage to the left posterior superior temporal gyrus is a significant predictor of auditory comprehension in chronic aphasia.

@Yourganov2016 · Study method: MLPA · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Yourganov2016**

▶ **Study context, limitations, review and source passages**

Phonological production 2 findings

Damage to left posterior perisylvian regions, including the posterior superior temporal gyrus, middle temporal gyrus, and Heschl's gyrus, is associated with impaired phonological processing in chronic stroke.

@Butler2014 · Study method: VBCM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Butler2014**

▶ **Study context, limitations, review and source passages**

Damage to the left posterior superior temporal gyrus is not significantly associated with phonological errors in object naming.

@Schwartz2012 · Study method: VLSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
No Association	**Strong**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Schwartz2012**

▶ **Study context, limitations, review and source passages**

Aphasia severity metric (instrument-agnostic) 1 finding

Damage to the left posterior superior temporal gyrus (Wernicke's area) is associated with language deficits in acute stroke.

@Corbetta2015 · Study method: MLPA · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Corbetta2015**

▶ **Study context, limitations, review and source passages**

Rhythm Production (Tapping) 1 finding

Damage to the left posterior superior temporal gyrus, extending into the supramarginal gyrus and arcuate fasciculus, is associated with impaired rhythm tapping in chronic aphasia.

@Kasdan2025 · Study method: VLSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Kasdan2025**

▶ **Study context, limitations, review and source passages**

Syntactic Processing Deficits in Aphasia 3 findings

Damage to the left posterior superior and middle temporal gyri is significantly associated with paragrammatism, independent of speech rate.

@Matchin2020 · Study method: LSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Strong**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Matchin2020**

▶ **Study context, limitations, review and source passages**

Damage to the left posterior superior temporal gyrus and inferior parietal regions is associated with agrammatic sentence comprehension (impaired performance on noncanonical versus canonical sentences).

@Rogalsky2018 · Study method: VLSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Strong**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Rogalsky2018**

▶ **Study context, limitations, review and source passages**

Extensive damage to the left posterior temporoparietal cortex, including the posterior superior temporal gyrus, is associated with ubiquitous and persistent deficits in sentence comprehension.

@Wilson2023 · Study method: LSM · longitudinal

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Strong**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Wilson2023**

▶ **Study context, limitations, review and source passages**

Fluency (speech fluency) 1 finding

Damage to the left posterior superior temporal gyrus is a significant predictor of speech fluency in chronic aphasia.

@Yourganov2016 · Study method: MLPA · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Yourganov2016**

▶ **Study context, limitations, review and source passages**

Speech repetition 1 finding

Damage to the left posterior superior temporal gyrus is a significant predictor of speech repetition in chronic aphasia.

@Yourganov2016 · Study method: MLPA · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Yourganov2016**

▶ **Study context, limitations, review and source passages**

Form-to-meaning processing (ventral stream) 1 finding

Damage to the left posterior superior temporal gyrus loads strongly on the form-to-meaning (ventral-stream) component, marking it as the second anchor of the ventral speech-processing stream alongside the posterior MTG.

@Fridriksson2018 · Study method: VLSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Strong**	**High**

Entry status: approved · Review provenance: legacy approval · Reviewer: auto-reviewer

▶ **Full citation for @Fridriksson2018**

▶ **Study context, limitations, review and source passages**

Phonological recognition 1 finding

Tissue concentration in the left posterior superior temporal gyrus uniquely covaries with the phonological recognition PCA factor in chronic post-stroke aphasia.

@Alyahya2018NounVerb · Study method: VBCM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: approved · Review provenance: legacy approval · Reviewer: auto-reviewer

▶ **Full citation for @Alyahya2018NounVerb**

▶ **Study context, limitations, review and source passages**

Left Posterior Planum Temporale (Area Spt) (left)

2.1% of the lesion · 100% of the region destroyed · 3 distinct findings

Atlas match includes a documented judgement call | Limited citation base: 2 citation keys

Also recorded in the knowledge base as: Left planum temporale

Conduction aphasia 2 findings

Damage to the left posterior planum temporale (area Spt) is associated with the symptoms of conduction aphasia, including impaired verbatim repetition and phonological short-term memory deficits.

@Buchsbaum2011 · Study method: LSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Buchsbaum2011**

▶ **Study context, limitations, review and source passages**

Area Spt in the left posterior planum temporale is recruited during both the encoding and delay phases of a phonological working memory task, behaviors that are impaired in conduction aphasia.

@Buchsbaum2011 · Study method: fMRI_activation · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Strong**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Buchsbaum2011**

▶ **Study context, limitations, review and source passages**

Phonological recognition 1 finding

Tissue concentration in the left planum temporale uniquely covaries with the phonological recognition PCA factor in chronic post-stroke aphasia (also implicated in phonological production).

@Alyahya2018NounVerb · Study method: VBCM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Strong**	**High**

Entry status: approved · Review provenance: legacy approval · Reviewer: auto-reviewer

▶ **Full citation for @Alyahya2018NounVerb**

▶ **Study context, limitations, review and source passages**

Left parietal operculum cortex (left)

2.3% of the lesion · 100% of the region destroyed · 1 distinct finding

Limited citation base: 1 citation key

Phonological production 1 finding

Tissue concentration in the left parietal operculum cortex uniquely covaries with the phonological production PCA factor in chronic post-stroke aphasia (the headline cluster of Factor 1 in the Discussion).

@Alyahya2018NounVerb · Study method: VBCM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Strong**	**High**

Entry status: approved · Review provenance: legacy approval · Reviewer: auto-reviewer

▶ **Full citation for @Alyahya2018NounVerb**

▶ **Study context, limitations, review and source passages**

Left Heschl's Gyrus (left)

1.3% of the lesion · 100% of the region destroyed · 5 distinct findings

Phonological recognition 2 findings

Damage to the left Heschl's gyrus is strongly associated with deficits in speech recognition.

@Thye2018 · Study method: VLSM, MLPA · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Strong**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Thye2018**

▶ **Study context, limitations, review and source passages**

Tissue concentration in the left Heschl's gyrus uniquely covaries with the phonological recognition PCA factor in chronic post-stroke aphasia (the headline cluster of Factor 4 in the Discussion).

@Alyahya2018NounVerb · Study method: VBCM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Strong**	**High**

Entry status: approved · Review provenance: legacy approval · Reviewer: auto-reviewer

▶ **Full citation for @Alyahya2018NounVerb**

▶ **Study context, limitations, review and source passages**

Broca's aphasia 1 finding

Damage to the left Heschl's gyrus contributes to Broca's aphasia classification (predictive relevance Z = 1.35).

@Yourganov2015Predicting · Study method: MLPA · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: approved · Review provenance: legacy approval · Reviewer: auto-reviewer

▶ **Full citation for @Yourganov2015Predicting**

▶ **Study context, limitations, review and source passages**

Conduction aphasia 1 finding

Damage to the left Heschl's gyrus contributes to conduction aphasia classification (predictive relevance Z = 1.47).

@Yourganov2015Predicting · Study method: MLPA · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: approved · Review provenance: human · Reviewer: michele

▶ **Full citation for @Yourganov2015Predicting**

▶ **Study context, limitations, review and source passages**

Wernicke's aphasia 1 finding

Damage to the left Heschl's gyrus contributes to Wernicke's aphasia classification (predictive relevance Z = 1.61).

@Yourganov2015Predicting · Study method: MLPA · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**Medium**

Entry status: approved · Review provenance: human · Reviewer: michele

▶ **Full citation for @Yourganov2015Predicting**

▶ **Study context, limitations, review and source passages**

Left rolandic operculum (left)

4.0% of the lesion · 100% of the region destroyed · 2 distinct findings

Limited citation base: 2 citation keys

Also recorded in the knowledge base as: Left central opercular cortex

Broca's aphasia 1 finding

Damage to the left rolandic operculum (the cortex superior and posterior to pars opercularis) is a strong predictor of Broca's aphasia (predictive relevance Z = 2.2).

@Yourganov2015Predicting · Study method: MLPA · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Strong**	**High**

Entry status: approved · Review provenance: legacy approval · Reviewer: auto-reviewer

▶ **Full citation for @Yourganov2015Predicting**

▶ **Study context, limitations, review and source passages**

Fluency (speech fluency) 1 finding

Tissue concentration in the left central opercular cortex uniquely covaries with the fluency PCA factor in chronic post-stroke aphasia (the strongest peak in the fluency cluster).

@Alyahya2018NounVerb · Study method: VBCM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Weak**	**High**

Entry status: approved · Review provenance: human · Reviewer: michele

▶ **Full citation for @Alyahya2018NounVerb**

▶ **Study context, limitations, review and source passages**

Left Superior Temporal Gyrus (Wernicke's Area and Planum Temporale) (left)

5.7% of the lesion · 100% of the region destroyed · 12 distinct findings

Atlas match includes a documented judgement call

Also recorded in the knowledge base as: Left Superior Temporal Gyrus (Wernicke's region, mid-posterior), Left Posterior Superior Temporal Gyrus and Sulcus (Auditory STM / Comprehension Shared Substrate)

Speech recognition phonological 1 finding

Lesions in the left superior temporal gyrus (including Wernicke's area) extending deep into the planum temporale, inferior to the Sylvian fissure, are associated with auditory-phonological speech recognition deficits in chronic post-stroke aphasia, as captured by a Speech Recognition factor loading on auditory lexical decision, phoneme discrimination, and rhyme discrimination tasks.

@Mirman2015 · Study method: VLSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Strong**	**High**

Entry status: approved · Review provenance: legacy approval · Reviewer: claude-cowork

▶ **Full citation for @Mirman2015**

▶ **Study context, limitations, review and source passages**

Semantics (semantic processing) 2 findings

Damage to the left posterior superior temporal gyrus (Wernicke's area) was not significantly associated with the production of semantic naming errors.

@Schwartz2009 · Study method: VLSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
No Association	**Moderate**	**High**

Entry status: approved · Review provenance: model · Reviewer: model:gemma4:26b · Review tier: A

▶ **Full citation for @Schwartz2009**

▶ **Study context, limitations, review and source passages**

In acute stroke, the severity of lexical-semantic impairment is strongly correlated with the magnitude of perfusion delay in left Brodmann's area 22 (Wernicke's area).

@Hillis2001 · Study method: PWI, behavioral_only · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Strong**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Hillis2001**

▶ **Study context, limitations, review and source passages**

Auditory comprehension 2 findings

Lesions to the left mid-posterior superior temporal gyrus (and adjacent STS and white matter) are associated with the comprehension deficits characterizing Wernicke's aphasia, with maximal lesion overlap in this region across a group of 12 WA patients.

@Robson2019 · Study method: LSM · longitudinal

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Weak**	**Medium**

Entry status: approved · Review provenance: human · Reviewer: michele

▶ **Full citation for @Robson2019**

▶ **Study context, limitations, review and source passages**

Language comprehension in Wernicke's aphasia shows strong and consistent recovery from 2.5 to 9 months post-onset, with significant improvement between each successive time point, but residual comprehension impairments remain in all participants at 9 MPO.

@Robson2019 · Study method: behavioral_only · longitudinal

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Weak**	**High**

Entry status: approved · Review provenance: human · Reviewer: michele

▶ **Full citation for @Robson2019**

▶ **Study context, limitations, review and source passages**

Wernicke's aphasia 1 finding

Damage to the left posterior and anterior temporal and parietal cortex is associated with Wernicke's aphasia.

@Henseler2014 · Study method: VLSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: approved · Review provenance: model · Reviewer: model:gemma4:26b · Review tier: A

▶ **Full citation for @Henseler2014**

▶ **Study context, limitations, review and source passages**

Auditory short term memory capacity 1 finding

Structural integrity of a focal region of the left posterior superior temporal gyrus and sulcus, lateral and posterior to Heschl's gyrus and overlapping with the inferior planum temporale, is causally necessary for auditory short-term memory capacity (digit span), even after controlling for speech production, speech perception, non-word repetition, verbal fluency, picture naming, and stroke volume.

@Leff2009 · Study method: VBCM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Strong**	**High**

Entry status: approved · Review provenance: legacy approval · Reviewer: claude-cowork

▶ **Full citation for @Leff2009**

▶ **Study context, limitations, review and source passages**

Spoken sentence comprehension 1 finding

Structural integrity of the left posterior superior temporal gyrus (same region supporting auditory STM capacity) also predicts spoken sentence comprehension ability, but NOT single spoken word comprehension or written sentence comprehension, consistent with a shared substrate for auditory STM and propositional speech comprehension.

@Leff2009 · Study method: VBCM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: approved · Review provenance: legacy approval · Reviewer: claude-cowork

▶ **Full citation for @Leff2009**

▶ **Study context, limitations, review and source passages**

Syntactic Processing Deficits in Aphasia 1 finding

Damage to the left posterior superior temporal gyrus (Wernicke's area) was not found to significantly affect sentence comprehension on the CYCLE-R.

@Dronkers2004 · Study method: VLSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
No Association	**Strong**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Dronkers2004**

▶ **Study context, limitations, review and source passages**

Pure word deafness 1 finding

Pure word-deafness results from the disconnection of Wernicke's area from auditory stimulation.

@Geschwind1965 · Study method: narrative_review · narrative-review

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Geschwind1965**

▶ **Study context, limitations, review and source passages**

Anomic aphasia 1 finding

Restored blood flow to left BA 22 (Wernicke's area) independently contributes to the prediction of early improvement in picture naming after stroke.

@Hillis2006 · Study method: LSM · longitudinal

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Hillis2006**

▶ **Study context, limitations, review and source passages**

Fluency (speech fluency) 1 finding

Damage to Wernicke's area is a significant predictor of speech production impairment.

@Hope2013 · Study method: MLPA · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Hope2013**

▶ **Study context, limitations, review and source passages**

Left Superior Temporal Gyrus (Dorsal Stream Node) (left)

4.5% of the lesion · 100% of the region destroyed · 12 distinct findings

Speech repetition 1 finding

The left posterior and anterior superior temporal gyrus is activated during sublexical speech repetition (pseudoword > word repetition), forming the temporal node of the dorsal language stream involved in sublexical speech perception and auditory-motor mapping.

@Saur2008 · Study method: fMRI_activation · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: approved · Review provenance: legacy approval · Reviewer: auto-reviewer

▶ **Full citation for @Saur2008**

▶ **Study context, limitations, review and source passages**

Form-to-meaning processing (ventral stream) 1 finding

Damage to the anterior and middle left superior temporal gyrus and left anterior IFG is associated with impaired language comprehension.

@Henseler2014 · Study method: VLSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: approved · Review provenance: model · Reviewer: model:gemma4:26b · Review tier: A

▶ **Full citation for @Henseler2014**

▶ **Study context, limitations, review and source passages**

Phonological recognition 2 findings

Damage to Heschl's gyrus is associated with impaired phonological recognition.

@Alyahya2020 · Study method: VBCM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Alyahya2020**

▶ **Study context, limitations, review and source passages**

The left mid-superior temporal gyrus is recruited for the processing of phonemes during auditory speech perception.

@DeWitt2012 · Study method: meta-analysis

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Strong**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @DeWitt2012**

▶ **Study context, limitations, review and source passages**

Semantics (semantic processing) 2 findings

Overall picture-naming performance on the BNT is associated with a network including the left superior temporal gyrus, though this association is reduced when controlling for speech production.

@Baldo2013 · Study method: VLSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Baldo2013**

▶ **Study context, limitations, review and source passages**

The left anterior superior temporal gyrus is recruited for auditory word-form recognition.

@DeWitt2012 · Study method: meta-analysis

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Strong**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @DeWitt2012**

▶ **Study context, limitations, review and source passages**

Syntactic Processing Deficits in Aphasia 1 finding

Damage to the left anterior superior temporal gyrus (BA22) is associated with deficits in the comprehension of simple sentences and basic morphosyntactic structures.

@Dronkers2004 · Study method: VLSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Strong**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Dronkers2004**

▶ **Study context, limitations, review and source passages**

Phonological production 1 finding

Damage to the left superior temporal gyrus is associated with impaired phonological skills in chronic stroke.

@Halai2017 · Study method: VBCM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Halai2017**

▶ **Study context, limitations, review and source passages**

Phonological dyslexia 1 finding

Damage to the left superior temporal gyrus is associated with phonological dyslexia.

@Rapcsak2009 · Study method: LSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Rapcsak2009**

▶ **Study context, limitations, review and source passages**

Aphasia severity metric (instrument-agnostic) 1 finding

Increased recruitment of left-hemispheric language networks (measured by activation volume index) is positively correlated with global language recovery after inhibitory rTMS to the right IFG.

@Thiel2013 · Study method: TMS, fMRI_activation · RCT

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Thiel2013**

▶ **Study context, limitations, review and source passages**

Broca's aphasia 1 finding

Damage to the left superior temporal gyrus contributes to Broca's aphasia classification in this multivariate model (predictive relevance Z = 1.99) — consistent with the view that classical Broca's requires damage extending into Wernicke's-area territory.

@Yourganov2015Predicting · Study method: MLPA · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: approved · Review provenance: legacy approval · Reviewer: auto-reviewer

▶ **Full citation for @Yourganov2015Predicting**

▶ **Study context, limitations, review and source passages**

Wernicke's aphasia 1 finding

Damage to the left superior temporal gyrus contributes to Wernicke's aphasia classification (predictive relevance Z = 1.31).

@Yourganov2015Predicting · Study method: MLPA · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**Medium**

Entry status: approved · Review provenance: human · Reviewer: michele

▶ **Full citation for @Yourganov2015Predicting**

▶ **Study context, limitations, review and source passages**

Left anterior middle temporal gyrus (left)

1.8% of the lesion · 99% of the region destroyed · 2 distinct findings

Limited citation base: 2 citation keys

Form-to-meaning processing (ventral stream) 1 finding

Damage to the left anterior middle temporal gyrus loads on the form-to-meaning (ventral-stream) component, consistent with its role as a semantic-hub region.

@Fridriksson2018 · Study method: VLSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: approved · Review provenance: legacy approval · Reviewer: auto-reviewer

▶ **Full citation for @Fridriksson2018**

▶ **Study context, limitations, review and source passages**

Semantics (semantic processing) 1 finding

Tissue concentration in the left anterior middle temporal gyrus uniquely covaries with the semantic PCA factor in chronic post-stroke aphasia, independent of phonological production, fluency, phonological recognition, and executive function components.

@Alyahya2018NounVerb · Study method: VBCM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Weak**	**High**

Entry status: approved · Review provenance: human · Reviewer: michele

▶ **Full citation for @Alyahya2018NounVerb**

▶ **Study context, limitations, review and source passages**

Left anterior supramarginal gyrus (left)

3.7% of the lesion · 97% of the region destroyed · 1 distinct finding

Limited citation base: 1 citation key

Form-to-articulation processing (dorsal stream) 1 finding

Damage to the anterior portion of the left supramarginal gyrus loads on the form-to-articulation (dorsal-stream) component, marking the inferior-parietal endpoint of the dorsal speech-processing stream.

@Fridriksson2018 · Study method: VLSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Strong**	**High**

Entry status: approved · Review provenance: legacy approval · Reviewer: auto-reviewer

▶ **Full citation for @Fridriksson2018**

▶ **Study context, limitations, review and source passages**

Left Arcuate Fasciculus and Superior Longitudinal Fascicle (Dorsal Language Pathway) (left)

1.1% of the lesion · 96% of the region destroyed · 5 distinct findings

Left posterior superior temporal gyrus 1 finding

The left arcuate fasciculus and superior longitudinal fascicle (AF/SLF) structurally connects the left posterior and anterior superior temporal gyrus to the left premotor cortex (pars opercularis BA44 and dorsal premotor BA6), forming the dorsal white-matter pathway subserving auditory-motor mapping during speech repetition.

@Saur2008 · Study method: DTI, fMRI_activation · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: approved · Review provenance: legacy approval · Reviewer: claude-cowork

▶ **Full citation for @Saur2008**

▶ **Study context, limitations, review and source passages**

Speech repetition 1 finding

The left dorsal pathway (AF/SLF connecting superior temporal gyrus to premotor cortex) is recruited during sublexical speech repetition, and disruption of this network after stroke is speculated to relate to conduction aphasia (impaired repetition with preserved comprehension and phonological paraphasias).

@Saur2008 · Study method: fMRI_activation, DTI · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Weak**	**Medium**

Entry status: approved · Review provenance: legacy approval · Reviewer: claude-cowork

▶ **Full citation for @Saur2008**

▶ **Study context, limitations, review and source passages**

Fluency (speech fluency) 2 findings

Damage to the left arcuate/superior longitudinal fasciculus in parietal white matter is associated with reduced speech fluency.

@Bates2003 · Study method: VLSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Strong**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Bates2003**

▶ **Study context, limitations, review and source passages**

Damage to the left superior longitudinal fasciculus is associated with deficits in both semantic and phonological verbal fluency.

@Li2017 · Study method: LSM, DTI · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Li2017**

▶ **Study context, limitations, review and source passages**

Phonological production 1 finding

Damage to the left dorsal superior longitudinal and arcuate fasciculus is associated with impaired speech repetition in acute stroke.

@Kummerer2013 · Study method: VLSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Strong**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Kummerer2013**

▶ **Study context, limitations, review and source passages**

Left Superior Longitudinal Fasciculus (left)

1.1% of the lesion · 96% of the region destroyed · 1 distinct finding

Limited citation base: 1 citation key

Phonological production 1 finding

Reduced integrity of the left superior longitudinal fasciculus is associated with impaired speech repetition in chronic stroke patients.

@Breier2008 · Study method: DTI · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Breier2008**

▶ **Study context, limitations, review and source passages**

Left precentral gyrus of the insula (left)

4.5% of the lesion · 96% of the region destroyed · 9 distinct findings

Atlas match includes a documented judgement call

Also recorded in the knowledge base as: Left Insula

Apraxia of speech 4 findings

Damage to the left precentral gyrus of the insula is necessary for apraxia of speech; all 25 patients with apraxia of speech had lesions encompassing this region, while none of 19 patients without apraxia of speech had lesions there.

@Dronkers1996 · Study method: LSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Strong**	**High**

Entry status: approved · Review provenance: legacy approval · Reviewer: auto-reviewer

▶ **Full citation for @Dronkers1996**

▶ **Study context, limitations, review and source passages**

Damage to the left superior precentral gyrus of the insula is associated with impaired articulation of complex words, specifically those that are multi-syllabic, require high articulatory travel, or contain initial consonant clusters.

@Baldo2011 · Study method: VLSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Baldo2011**

▶ **Study context, limitations, review and source passages**

Damage to the left superior precentral gyrus of the insula does not significantly impair the articulation of simple, single-syllable words with minimal articulatory travel and no initial consonant clusters.

@Baldo2011 · Study method: VLSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
No Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Baldo2011**

▶ **Study context, limitations, review and source passages**

Damage or dysfunction of the left insula, including the anterior insula and the superior tip of the precentral gyrus of the insula, is not associated with apraxia of speech in acute stroke.

@Hillis2004 · Study method: LSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
No Association	**Strong**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Hillis2004**

▶ **Study context, limitations, review and source passages**

Form-to-articulation processing (dorsal stream) 1 finding

Damage to the left insula and adjacent opercular cortex is associated with impaired articulation and prosody in spontaneous speech.

@Henseler2014 · Study method: VLSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: approved · Review provenance: model · Reviewer: model:gemma4:26b · Review tier: A

▶ **Full citation for @Henseler2014**

▶ **Study context, limitations, review and source passages**

Phonological dyslexia 1 finding

Damage to the left insula is associated with phonological dyslexia.

@Rapcsak2009 · Study method: LSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Rapcsak2009**

▶ **Study context, limitations, review and source passages**

Fluency (speech fluency) 2 findings

Damage to the left anterior insula is strongly associated with reduced speech fluency.

@Bates2003 · Study method: VLSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Strong**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Bates2003**

▶ **Study context, limitations, review and source passages**

The anterior insula is critical for speech fluency independently of damage to Broca's area.

@Bates2003 · Study method: VLSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Strong**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Bates2003**

▶ **Study context, limitations, review and source passages**

Global aphasia 1 finding

Damage to the left insula contributes to global aphasia classification (predictive relevance Z = 1.66).

@Yourganov2015Predicting · Study method: MLPA · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: approved · Review provenance: legacy approval · Reviewer: auto-reviewer

▶ **Full citation for @Yourganov2015Predicting**

▶ **Study context, limitations, review and source passages**

Left arcuate fasciculus (left)

5.0% of the lesion · 90% of the region destroyed · 21 distinct findings

Atlas match includes a documented judgement call

Also recorded in the knowledge base as: Left posterior arcuate fasciculus, Left long segment arcuate fasciculus

Speech production rate 1 finding

Lesion load of the left arcuate fasciculus (proportion of the tract damaged by a patient's stroke) significantly predicts rate of speech (words per minute), whereas overall lesion size, extreme capsule lesion load, and uncinate fasciculus lesion load do not.

@Marchina2011 · Study method: tractography · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: approved · Review provenance: legacy approval · Reviewer: claude-cowork

▶ **Full citation for @Marchina2011**

▶ **Study context, limitations, review and source passages**

Speech informativeness 1 finding

Lesion load of the left arcuate fasciculus significantly predicts informativeness of speech (%CIUs — correct information units as a proportion of total words), with greater tract damage associated with lower informativeness. Overall lesion size, extreme capsule, and uncinate fasciculus lesion loads are not significant predictors.

@Marchina2011 · Study method: tractography · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: approved · Review provenance: legacy approval · Reviewer: claude-cowork

▶ **Full citation for @Marchina2011**

▶ **Study context, limitations, review and source passages**

Speech efficiency 1 finding

Lesion load of the left arcuate fasciculus is the strongest predictor of overall speech efficiency (CIUs per minute), explaining 45% of unique variance in a regression model that also includes overall lesion size and extreme capsule and uncinate fasciculus lesion loads.

@Marchina2011 · Study method: tractography · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: approved · Review provenance: legacy approval · Reviewer: claude-cowork

▶ **Full citation for @Marchina2011**

▶ **Study context, limitations, review and source passages**

Confrontation naming 1 finding

Lesion load of the left arcuate fasciculus significantly predicts confrontation naming ability (Boston Naming Test score), whereas extreme capsule lesion load and overall lesion size do not; uncinate fasciculus lesion load shows a nonsignificant trend.

@Marchina2011 · Study method: tractography · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: approved · Review provenance: legacy approval · Reviewer: claude-cowork

▶ **Full citation for @Marchina2011**

▶ **Study context, limitations, review and source passages**

Left Extreme Capsule (Ventral Language Pathway) 1 finding

Extreme capsule lesion load does not significantly predict any measure of speech production (rate, informativeness, efficiency) or naming ability, contrary to predictions from dual-stream models in which the extreme capsule mediates sound-to-meaning mapping.

@Marchina2011 · Study method: tractography · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
No Association	**Weak**	**High**

Entry status: approved · Review provenance: legacy approval · Reviewer: claude-cowork

▶ **Full citation for @Marchina2011**

▶ **Study context, limitations, review and source passages**

Phonological production 6 findings

Reduced integrity of the left arcuate fasciculus is associated with impaired speech repetition in chronic stroke patients.

@Breier2008 · Study method: DTI · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Breier2008**

▶ **Study context, limitations, review and source passages**

Damage to the left arcuate fasciculus is associated with impaired phonological processing in chronic stroke.

@Butler2014 · Study method: VBCM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Butler2014**

▶ **Study context, limitations, review and source passages**

The superior longitudinal fasciculus/arcuate fasciculus (SLF/AF) is involved in phonological processing during speech perception and production.

@Dick2012 · Study method: narrative_review · narrative-review

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Weak**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Dick2012**

▶ **Study context, limitations, review and source passages**

Integrity of the left arcuate fasciculus is significantly related to language production in chronic aphasia.

@Ivanova2016 · Study method: DTI · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Ivanova2016**

▶ **Study context, limitations, review and source passages**

Damage to the anterior portion of the left arcuate fasciculus is associated with increased phonological errors in object naming.

@Schwartz2012 · Study method: VLSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Schwartz2012**

▶ **Study context, limitations, review and source passages**

Tissue concentration in the posterior segment of the left arcuate fasciculus uniquely covaries with the phonological production PCA factor in chronic post-stroke aphasia, after partialling out the four other principal components.

@Alyahya2018NounVerb · Study method: VBCM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Strong**	**High**

Entry status: approved · Review provenance: legacy approval · Reviewer: auto-reviewer

▶ **Full citation for @Alyahya2018NounVerb**

▶ **Study context, limitations, review and source passages**

Form-to-meaning processing (ventral stream) 1 finding

Reduced integrity of the left arcuate fasciculus is marginally associated with comprehension deficits, though this relationship depends on cortical damage.

@Breier2008 · Study method: DTI · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Weak**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Breier2008**

▶ **Study context, limitations, review and source passages**

Left arcuate fasciculus 2 findings

The left arcuate fasciculus consists of a direct segment connecting Wernicke's and Broca's areas, and an indirect pathway comprising anterior and posterior segments via the inferior parietal lobule.

@Catani2008 · Study method: narrative_review · narrative-review

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Catani2008**

▶ **Study context, limitations, review and source passages**

Extreme left lateralization of the direct arcuate fasciculus segment occurs in approximately 60% of the normal population, with higher prevalence in males (85%) than females (40%).

@Catani2008 · Study method: narrative_review · narrative-review

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Catani2008**

▶ **Study context, limitations, review and source passages**

Speech repetition 1 finding

Damage to the posterior portion of the left arcuate fasciculus is strongly associated with impaired speech repetition.

@Fridriksson2010 · Study method: VLSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Fridriksson2010**

▶ **Study context, limitations, review and source passages**

Fluency (speech fluency) 1 finding

Damage to the superior longitudinal fasciculus is a significant predictor of speech production impairment.

@Hope2013 · Study method: MLPA · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Hope2013**

▶ **Study context, limitations, review and source passages**

Aphasia 2 findings

Loss of leftward asymmetry in the number of arcuate fasciculus fibers predicts the presence of aphasia at discharge in acute left MCA stroke patients.

@Hosomi2009 · Study method: tractography · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Weak**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Hosomi2009**

▶ **Study context, limitations, review and source passages**

Lesions of the left arcuate fasciculus (>15% overlap) were not significantly associated with the occurrence of aphasia in acute stroke patients without cortical language area lesions.

@Oeckel2021 · Study method: LSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
No Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Oeckel2021**

▶ **Study context, limitations, review and source passages**

Aphasia severity metric (instrument-agnostic) 1 finding

Damage to the posterior and long segments of the left arcuate fasciculus is associated with poorer chronic language outcomes.

@Ramsey2017 · Study method: MLPA · longitudinal

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Ramsey2017**

▶ **Study context, limitations, review and source passages**

Broca's aphasia 1 finding

Damage to the long segment of the left arcuate fasciculus is among the strongest predictors of Broca's aphasia (predictive relevance Z = 2.85).

@Yourganov2015Predicting · Study method: MLPA · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Strong**	**High**

Entry status: approved · Review provenance: legacy approval · Reviewer: auto-reviewer

▶ **Full citation for @Yourganov2015Predicting**

▶ **Study context, limitations, review and source passages**

Conduction aphasia 1 finding

Damage to the posterior segment of the left arcuate fasciculus is the strongest predictor of conduction aphasia in a multivariate SVM classification of 98 chronic stroke aphasia patients (predictive relevance Z = 2.48).

@Yourganov2015Predicting · Study method: MLPA · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Strong**	**High**

Entry status: approved · Review provenance: human · Reviewer: michele

▶ **Full citation for @Yourganov2015Predicting**

▶ **Study context, limitations, review and source passages**

Left Supramarginal Gyrus (left)

7.1% of the lesion · 88% of the region destroyed · 4 distinct findings

Speech repetition 1 finding

Hypoperfusion in the inferior portion of the left supramarginal gyrus is a strong predictor of impaired speech repetition.

@Fridriksson2010 · Study method: VBCM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Fridriksson2010**

▶ **Study context, limitations, review and source passages**

Phonological dyslexia 1 finding

Damage to the left supramarginal gyrus is associated with phonological dyslexia.

@Rapcsak2009 · Study method: LSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Rapcsak2009**

▶ **Study context, limitations, review and source passages**

Phonological production 1 finding

Damage to the left supramarginal gyrus is associated with increased phonological errors in object naming.

@Schwartz2012 · Study method: VLSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Strong**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Schwartz2012**

▶ **Study context, limitations, review and source passages**

Broca's aphasia 1 finding

Damage to the left supramarginal gyrus contributes to Broca's aphasia classification (predictive relevance Z = 1.66).

@Yourganov2015Predicting · Study method: MLPA · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: approved · Review provenance: legacy approval · Reviewer: auto-reviewer

▶ **Full citation for @Yourganov2015Predicting**

▶ **Study context, limitations, review and source passages**

Left Temporoparietal Junction (left)

9.8% of the lesion · 85% of the region destroyed · 3 distinct findings

Atlas match includes a documented judgement call | Limited citation base: 2 citation keys

Prognosis 2 findings

Damage to the left temporoparietal white matter (TPWM) is associated with poor 3-month functional outcome.

@Benghanem2019 · Study method: LSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Benghanem2019**

▶ **Study context, limitations, review and source passages**

The predictive value of left temporoparietal junction damage on outcome is significant only in patients with initially severe aphasia.

@Benghanem2019 · Study method: LSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Benghanem2019**

▶ **Study context, limitations, review and source passages**

Speech repetition 1 finding

Hypoperfusion in the left temporoparietal junction (TPJ) is associated with impaired speech repetition.

@Fridriksson2010 · Study method: VBCM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Fridriksson2010**

▶ **Study context, limitations, review and source passages**

Left anterior arcuate fasciculus (left)

7.3% of the lesion · 84% of the region destroyed · 5 distinct findings

Atlas match includes a documented judgement call

Fluency (speech fluency) 3 findings

Damage to the anterior segment of the left arcuate fasciculus is a robust predictor of impaired speech fluency in aphasia.

@Fridriksson2013 · Study method: VBCM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Strong**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Fridriksson2013**

▶ **Study context, limitations, review and source passages**

Damage to the anterior segment of the left arcuate fasciculus is the sole significant predictor of long-term speech production impairment when controlling for other left frontal regions.

@GajardoVidal2021 · Study method: RLSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Strong**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @GajardoVidal2021**

▶ **Study context, limitations, review and source passages**

Tissue concentration in the anterior segment of the left arcuate fasciculus uniquely covaries with the fluency PCA factor in chronic post-stroke aphasia, after partialling out the four other principal components.

@Alyahya2018NounVerb · Study method: VBCM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: approved · Review provenance: human · Reviewer: michele

▶ **Full citation for @Alyahya2018NounVerb**

▶ **Study context, limitations, review and source passages**

Broca's aphasia 1 finding

Damage to the anterior segment of the left arcuate fasciculus is the strongest predictor of Broca's aphasia in a multivariate (SVM) classification of 98 chronic stroke aphasia patients (predictive relevance Z = 3.2).

@Yourganov2015Predicting · Study method: MLPA · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Strong**	**High**

Entry status: approved · Review provenance: legacy approval · Reviewer: auto-reviewer

▶ **Full citation for @Yourganov2015Predicting**

▶ **Study context, limitations, review and source passages**

Global aphasia 1 finding

Damage to the anterior segment of the left arcuate fasciculus contributes to global aphasia classification (predictive relevance Z = 1.65).

@Yourganov2015Predicting · Study method: MLPA · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: approved · Review provenance: legacy approval · Reviewer: auto-reviewer

▶ **Full citation for @Yourganov2015Predicting**

▶ **Study context, limitations, review and source passages**

Left Anterior Temporal Lobe (left)

10.1% of the lesion · 83% of the region destroyed · 8 distinct findings

Atlas match includes a documented judgement call

Semantic errors in naming 1 finding

Focal lesions of the left anterior temporal lobe (superior and middle portions) are associated with increased production of semantic errors in picture naming in patients with post-stroke aphasia, independently of multimodal semantic recognition ability.

@Mirman2015 · Study method: VLSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Strong**	**High**

Entry status: approved · Review provenance: legacy approval · Reviewer: auto-reviewer

▶ **Full citation for @Mirman2015**

▶ **Study context, limitations, review and source passages**

Syntactic Processing Deficits in Aphasia 1 finding

The left anterior temporal lobe is implicated in general sentence processing, but is not strongly associated with the specific agrammatic comprehension pattern (noncanonical < canonical).

@Rogalsky2018 · Study method: VLSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Mixed Results	**Moderate**	**High**

Entry status: approved · Review provenance: model · Reviewer: model:gemma4:26b · Review tier: A

▶ **Full citation for @Rogalsky2018**

▶ **Study context, limitations, review and source passages**

Semantics (semantic processing) 5 findings

Damage to the left anterior temporal fusiform cortex is associated with impaired semantic processing.

@Alyahya2020 · Study method: VBCM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Alyahya2020**

▶ **Study context, limitations, review and source passages**

Damage to the left anterior temporal lobe, specifically the anterior middle temporal gyrus and temporal stem, is associated with impaired semantic processing in chronic stroke.

@Butler2014 · Study method: VBCM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Butler2014**

▶ **Study context, limitations, review and source passages**

Damage to the left anterior temporal lobe is associated with impaired semantic ability in chronic stroke.

@Halai2017 · Study method: VBCM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Halai2017**

▶ **Study context, limitations, review and source passages**

Damage to the left anterior temporal lobe is associated with an increased rate of semantic naming errors.

@Schwartz2009 · Study method: VLSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Strong**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Schwartz2009**

▶ **Study context, limitations, review and source passages**

The association between the left anterior temporal lobe and semantic naming errors persists after controlling for non-verbal semantic comprehension, suggesting a specific role in the production stage of lexical access.

@Schwartz2009 · Study method: VLSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Strong**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Schwartz2009**

▶ **Study context, limitations, review and source passages**

Anomic aphasia 1 finding

Naming of compound nouns correlates with the integrity of the left anterior temporal lobe.

@Henseler2014 · Study method: VLSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Henseler2014**

▶ **Study context, limitations, review and source passages**

Left posterior supramarginal gyrus (left)

3.4% of the lesion · 80% of the region destroyed · 2 distinct findings

Limited citation base: 2 citation keys

Form-to-meaning processing (ventral stream) 1 finding

Damage to the posterior portion of the left supramarginal gyrus loads on the form-to-meaning (ventral-stream) component, contrasting with the anterior SMG that loads on the dorsal stream.

@Fridriksson2018 · Study method: VLSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Strong**	**High**

Entry status: approved · Review provenance: legacy approval · Reviewer: auto-reviewer

▶ **Full citation for @Fridriksson2018**

▶ **Study context, limitations, review and source passages**

Phonological production 1 finding

Tissue concentration in the left posterior supramarginal gyrus uniquely covaries with the phonological production PCA factor in chronic post-stroke aphasia.

@Alyahya2018NounVerb · Study method: VBCM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: approved · Review provenance: legacy approval · Reviewer: auto-reviewer

▶ **Full citation for @Alyahya2018NounVerb**

▶ **Study context, limitations, review and source passages**

Left Superior Temporal Sulcus (left)

6.8% of the lesion · 79% of the region destroyed · 2 distinct findings

Atlas match includes a documented judgement call

Limited citation base: 2 citation keys

Semantics (semantic processing) 1 finding

The left anterior superior temporal sulcus is specifically recruited during the processing of verbally encoded semantic knowledge.

@Binder2009 · Study method: meta-analysis · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Strong**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Binder2009**

▶ **Study context, limitations, review and source passages**

Auditory comprehension 1 finding

The left anterior superior temporal sulcus is recruited for the processing of phrase-length auditory stimuli.

@DeWitt2012 · Study method: meta-analysis

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Strong**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @DeWitt2012**

▶ **Study context, limitations, review and source passages**

Left temporal pole (left)

7.4% of the lesion · 78% of the region destroyed · 1 distinct finding

Limited citation base: 1 citation key

Semantics (semantic processing) 1 finding

Tissue concentration in the left temporal pole uniquely covaries with the semantic PCA factor in chronic post-stroke aphasia.

@Alyahya2018NounVerb · Study method: VBCM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Weak**	**High**

Entry status: approved · Review provenance: human · Reviewer: michele

▶ **Full citation for @Alyahya2018NounVerb**

▶ **Study context, limitations, review and source passages**

Left Angular Gyrus (left)

2.9% of the lesion · 77% of the region destroyed · 12 distinct findings

Object naming 1 finding

Greater damage to the left angular gyrus is independently associated with worse object naming (BNT) performance in acute left hemisphere stroke.

@Breining2022 · Study method: MLPA · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: approved · Review provenance: human · Reviewer: michele

▶ **Full citation for @Breining2022**

▶ **Study context, limitations, review and source passages**

Action naming 1 finding

Greater damage to the left angular gyrus is associated with worse action naming (HANA) performance in acute left hemisphere stroke; the effect is not independently statistically significant but the region appears in the LASSO-selected model.

@Breining2022 · Study method: MLPA · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Mixed Results	**Weak**	**High**

Entry status: approved · Review provenance: human · Reviewer: michele

▶ **Full citation for @Breining2022**

▶ **Study context, limitations, review and source passages**

Phonological production 2 findings

Damage to the left angular gyrus is associated with impaired phonological production.

@Alyahya2020 · Study method: VBCM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Alyahya2020**

▶ **Study context, limitations, review and source passages**

Tissue concentration in the left angular gyrus uniquely covaries with the phonological production PCA factor in chronic post-stroke aphasia.

@Alyahya2018NounVerb · Study method: VBCM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: approved · Review provenance: legacy approval · Reviewer: auto-reviewer

▶ **Full citation for @Alyahya2018NounVerb**

▶ **Study context, limitations, review and source passages**

Semantics (semantic processing) 2 findings

The left angular gyrus is a primary hub of the semantic processing network in healthy adults.

@Binder2009 · Study method: meta-analysis · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Strong**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Binder2009**

▶ **Study context, limitations, review and source passages**

The left angular gyrus is specifically recruited during the processing of perceptually encoded (concrete) knowledge.

@Binder2009 · Study method: meta-analysis · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Strong**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Binder2009**

▶ **Study context, limitations, review and source passages**

Syntactic Processing Deficits in Aphasia 1 finding

Damage to the left superior temporal sulcus and angular gyrus (BA39) is associated with deficits in sentence comprehension, potentially mediated by auditory short-term memory impairments.

@Dronkers2004 · Study method: VLSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Strong**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Dronkers2004**

▶ **Study context, limitations, review and source passages**

Cross modal association 1 finding

The left angular gyrus is a specialized region for forming and storing cross-modal associations, particularly between visual and auditory modalities.

@Geschwind1965 · Study method: narrative_review · narrative-review

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Geschwind1965**

▶ **Study context, limitations, review and source passages**

Pure alexia without agraphia 1 finding

Pure word-blindness without agraphia results from the disconnection of the intact right visual cortex from the left angular gyrus.

@Geschwind1965 · Study method: narrative_review · narrative-review

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Geschwind1965**

▶ **Study context, limitations, review and source passages**

Alexia 1 finding

The left angular gyrus is a critical region for written language after controlling for verbal working memory.

@Henseler2014 · Study method: VLSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Henseler2014**

▶ **Study context, limitations, review and source passages**

Form-to-meaning processing (ventral stream) 1 finding

Damage to the left angular gyrus loads on the form-to-meaning (ventral-stream) component.

@Fridriksson2018 · Study method: VLSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: approved · Review provenance: legacy approval · Reviewer: auto-reviewer

▶ **Full citation for @Fridriksson2018**

▶ **Study context, limitations, review and source passages**

Wernicke's aphasia 1 finding

Damage to the left angular gyrus is the strongest predictor of Wernicke's aphasia in a multivariate SVM classification (predictive relevance Z = 1.76).

@Yourganov2015Predicting · Study method: MLPA · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**Medium**

Entry status: approved · Review provenance: human · Reviewer: michele

▶ **Full citation for @Yourganov2015Predicting**

▶ **Study context, limitations, review and source passages**

Left temporal pole MTG (left)

2.3% of the lesion · 76% of the region destroyed · 2 distinct findings

Limited citation base: 1 citation key

Global aphasia 1 finding

Damage to the left temporal pole (middle temporal gyrus subdivision) contributes to global aphasia classification (predictive relevance Z = 2.15).

@Yourganov2015Predicting · Study method: MLPA · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Strong**	**High**

Entry status: approved · Review provenance: legacy approval · Reviewer: auto-reviewer

▶ **Full citation for @Yourganov2015Predicting**

▶ **Study context, limitations, review and source passages**

Wernicke's aphasia 1 finding

Damage to the left temporal pole (middle temporal gyrus subdivision) contributes to Wernicke's aphasia classification (predictive relevance Z = 1.47).

@Yourganov2015Predicting · Study method: MLPA · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Weak**	**Medium**

Entry status: approved · Review provenance: human · Reviewer: michele

▶ **Full citation for @Yourganov2015Predicting**

▶ **Study context, limitations, review and source passages**

Left Anterior Temporal Lobe (Residual Comprehension Substrate in Wernicke's Aphasia) (left)

10.2% of the lesion · 75% of the region destroyed · 2 distinct findings

Atlas match includes a documented judgement call | Limited citation base: 1 citation key

Semantic comprehension residual 1 finding

In chronic Wernicke's aphasia (left posterior temporoparietal lesions), the left anterior temporal lobe — including the anterior fusiform gyrus, temporal pole, and anterior middle temporal gyrus — shows significantly greater activation than controls during semantic animate-inanimate judgements of pictures and written words, indicating that ATL regions become the primary substrate supporting residual multimodal semantic comprehension when posterior temporal resources are damaged.

@Robson2014 · Study method: fMRI_activation · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: approved · Review provenance: legacy approval · Reviewer: claude-cowork

▶ **Full citation for @Robson2014**

▶ **Study context, limitations, review and source passages**

Right anterior superior temporal gyrus 1 finding

Written word semantic processing in Wernicke's aphasia — but not picture processing — additionally recruits the right anterior superior temporal gyrus and sulcus, a region associated with recovery from auditory-verbal comprehension impairment. This suggests right ATL recruitment is specifically linked to residual phonological pathway engagement during written word processing.

@Robson2014 · Study method: fMRI_activation · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Weak**	**Medium**

Entry status: approved · Review provenance: legacy approval · Reviewer: claude-cowork

▶ **Full citation for @Robson2014**

▶ **Study context, limitations, review and source passages**

Left Middle Longitudinal Fasciculus (left)

3.9% of the lesion · 72% of the region destroyed · 1 distinct finding

Limited citation base: 1 citation key

Semantics (semantic processing) 1 finding

The middle longitudinal fasciculus is part of the ventral sound-to-meaning pathway.

@Dick2012 · Study method: narrative_review · narrative-review

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Weak**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Dick2012**

▶ **Study context, limitations, review and source passages**

Left temporal pole STG (left)

3.7% of the lesion · 71% of the region destroyed · 1 distinct finding

Limited citation base: 1 citation key

Wernicke's aphasia 1 finding

Damage to the left temporal pole (superior temporal gyrus subdivision) contributes to Wernicke's aphasia classification (predictive relevance Z = 1.34).

@Yourganov2015Predicting · Study method: MLPA · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Weak**	**Medium**

Entry status: approved · Review provenance: human · Reviewer: michele

▶ **Full citation for @Yourganov2015Predicting**

▶ **Study context, limitations, review and source passages**

Left Temporal Lobe (left)

23.9% of the lesion · 67% of the region destroyed · 1 distinct finding

Limited citation base: 1 citation key

Anomic aphasia 1 finding

Higher nodal betweenness centrality of the left temporal lobe is associated with greater improvement in naming following speech therapy.

@Bonilha2016 · Study method: DTI · longitudinal

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Bonilha2016**

▶ **Study context, limitations, review and source passages**

Left Middle Temporal Gyrus (Ventral Stream Node) (left) — 6.8% of the lesion · 66% of the region destroyed · 11 distinct findings

Auditory comprehension 1 finding

The left posterior and anterior middle temporal gyrus is activated during sentence-level auditory comprehension (listening to meaningful vs. pseudo sentences), forming the temporal node of the ventral language stream involved in lexical-semantic and conceptual processing.

@Saur2008 · Study method: fMRI_activation · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: approved · Review provenance: legacy approval · Reviewer: auto-reviewer

▶ **Full citation for @Saur2008**

▶ **Study context, limitations, review and source passages**

Anomic aphasia 3 findings

Integrity of the posterior middle temporal gyrus and left orbital IFG is critical for picture naming.

@Henseler2014 · Study method: VLSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: approved · Review provenance: model · Reviewer: model:gemma4:26b · Review tier: A

▶ **Full citation for @Henseler2014**

▶ **Study context, limitations, review and source passages**

Damage to the posterior portion of the left middle temporal lobe and the temporal-occipital junction is associated with a reduced likelihood of improvement in naming following treatment.

@Fridriksson2010Anomia · Study method: LSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Inverse Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Fridriksson2010Anomia**

▶ **Study context, limitations, review and source passages**

Restored blood flow to the left posterior middle temporal/fusiform gyrus (BA 37) is strongly associated with acute improvement in picture naming after stroke.

@Hillis2006 · Study method: LSM · longitudinal

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Strong**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Hillis2006**

▶ **Study context, limitations, review and source passages**

Semantics (semantic processing) 7 findings

Damage to the left posterior middle temporal gyrus is associated with impaired semantic processing.

@Alyahya2020 · Study method: VBCM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Alyahya2020**

▶ **Study context, limitations, review and source passages**

Damage to the left mid-posterior middle temporal gyrus is critically associated with the ability to retrieve a name associated with an object (lexical-semantic retrieval).

@Baldo2013 · Study method: VLSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Strong**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Baldo2013**

▶ **Study context, limitations, review and source passages**

Damage to the left middle temporal gyrus is the most reliable predictor of impaired auditory comprehension.

@Bates2003 · Study method: VLSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Strong**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Bates2003**

▶ **Study context, limitations, review and source passages**

The middle temporal gyrus is a significant factor in auditory comprehension independently of Wernicke's area.

@Bates2003 · Study method: VLSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Strong**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Bates2003**

▶ **Study context, limitations, review and source passages**

The left middle temporal gyrus is a primary hub of the semantic processing network in healthy adults.

@Binder2009 · Study method: meta-analysis · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Strong**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Binder2009**

▶ **Study context, limitations, review and source passages**

Damage to the left posterior middle temporal gyrus and underlying white matter is associated with deficits in word-level language comprehension.

@Dronkers2004 · Study method: VLSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Strong**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Dronkers2004**

▶ **Study context, limitations, review and source passages**

Damage to the left anterior to mid middle temporal gyrus is associated with an increased rate of semantic naming errors.

@Schwartz2009 · Study method: VLSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Strong**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Schwartz2009**

▶ **Study context, limitations, review and source passages**

Left posterior middle temporal gyrus (left)

3.3% of the lesion · 64% of the region destroyed · 1 distinct finding

Limited citation base: 1 citation key

Form-to-meaning processing (ventral stream) 1 finding

Damage to the left posterior middle temporal gyrus loads strongly on the form-to-meaning (ventral-stream) processing component in chronic post-stroke aphasia, identifying it as the primary anchor of the ventral speech-processing stream.

@Fridriksson2018 · Study method: VLSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Strong**	**High**

Entry status: approved · Review provenance: legacy approval · Reviewer: auto-reviewer

▶ **Full citation for @Fridriksson2018**

▶ **Study context, limitations, review and source passages**

Left putamen (left)

2.5% of the lesion · 62% of the region destroyed · 2 distinct findings

Limited citation base: 1 citation key

Global aphasia 1 finding

Damage to the left putamen contributes to global aphasia classification (predictive relevance Z = 2.02).

@Yourganov2015Predicting · Study method: MLPA · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Strong**	**High**

Entry status: approved · Review provenance: legacy approval · Reviewer: auto-reviewer

▶ **Full citation for @Yourganov2015Predicting**

▶ **Study context, limitations, review and source passages**

Wernicke's aphasia 1 finding

Damage to the left putamen contributes to Wernicke's aphasia classification (predictive relevance Z = 1.3).

@Yourganov2015Predicting · Study method: MLPA · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Weak**	**Medium**

Entry status: approved · Review provenance: human · Reviewer: michele

▶ **Full citation for @Yourganov2015Predicting**

▶ **Study context, limitations, review and source passages**

Left IFG Pars Opercularis (Broca's area, dorsal stream hub) (left)

1.8% of the lesion · 60% of the region destroyed · 24 distinct findings

Also recorded in the knowledge base as: Inferior Frontal Gyrus, pars opercularis

Speech fluency 1 finding

Damage to left IFG pars opercularis is the strongest independent predictor of impaired speech fluency in chronic left-hemisphere stroke aphasia, accounting for 40% of variance in a multivariate region-wise LSM analysis.

@Fridriksson2018Anatomy · Study method: LSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Strong**	**High**

Entry status: approved · Review provenance: legacy approval · Reviewer: auto-reviewer

▶ **Full citation for @Fridriksson2018Anatomy**

▶ **Study context, limitations, review and source passages**

Aphasia quotient 1 finding

Damage to left IFG pars opercularis contributes to overall aphasia severity (WAB Aphasia Quotient), though pars opercularis is less strongly dominant than STG in the univariate analysis; aphasia severity involves an extensive cortical network primarily within the dorsal stream.

@Fridriksson2018Anatomy · Study method: LSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Strong**	**High**

Entry status: approved · Review provenance: legacy approval · Reviewer: auto-reviewer

▶ **Full citation for @Fridriksson2018Anatomy**

▶ **Study context, limitations, review and source passages**

Letter fluency 1 finding

Damage to left frontal cortex including BA 44 (pars opercularis) is associated with impaired letter fluency performance in chronic left-hemisphere stroke patients.

@Baldo2006 · Study method: VLSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: approved · Review provenance: legacy approval · Reviewer: auto-reviewer

▶ **Full citation for @Baldo2006**

▶ **Study context, limitations, review and source passages**

Category fluency 1 finding

Damage to left temporal cortex (BA 22, 37, 41, 42) is associated with impaired category fluency in chronic left-hemisphere stroke patients, whereas frontal damage is less specifically associated with category fluency deficits.

@Baldo2006 · Study method: VLSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: approved · Review provenance: legacy approval · Reviewer: auto-reviewer

▶ **Full citation for @Baldo2006**

▶ **Study context, limitations, review and source passages**

Fluency (speech fluency) 4 findings

Damage to Broca's area is not especially important for speech fluency once lesions to the insula have been accounted for.

@Bates2003 · Study method: VLSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
No Association	**Moderate**	**High**

Entry status: approved · Review provenance: model · Reviewer: model:gemma4:26b · Review tier: A

▶ **Full citation for @Bates2003**

▶ **Study context, limitations, review and source passages**

Damage to the left inferior frontal gyrus pars opercularis is a significant predictor of speech fluency in chronic aphasia.

@Yourganov2016 · Study method: MLPA · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: approved · Review provenance: model · Reviewer: model:gemma4:26b · Review tier: A

▶ **Full citation for @Yourganov2016**

▶ **Study context, limitations, review and source passages**

Damage to the left IFG pars opercularis (BA44) does not uniquely contribute to long-term speech production outcome when damage to the anterior arcuate fasciculus is controlled.

@GajardoVidal2021 · Study method: RLSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
No Association	**Strong**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @GajardoVidal2021**

▶ **Study context, limitations, review and source passages**

Damage to the parietal operculum is a significant predictor of speech production impairment.

@Hope2013 · Study method: MLPA · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Hope2013**

▶ **Study context, limitations, review and source passages**

Phonological production 1 finding

Damage to the left inferior frontal gyrus is associated with impaired phonological processing, although the relationship is weaker than for posterior perisylvian regions.

@Butler2014 · Study method: VBCM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Weak**	**High**

Entry status: approved · Review provenance: model · Reviewer: model:gemma4:26b · Review tier: A

▶ **Full citation for @Butler2014**

▶ **Study context, limitations, review and source passages**

Broca's aphasia 5 findings

Damage to the left inferior frontal gyrus, including Broca's area, is associated with Broca's aphasia.

@Henseler2014 · Study method: VLSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: approved · Review provenance: model · Reviewer: model:gemma4:26b · Review tier: A

▶ **Full citation for @Henseler2014**

▶ **Study context, limitations, review and source passages**

In acute stroke, ischemia involving any part of Broca area (BA 44/45) is strongly associated with Broca or global aphasia.

@Ochfeld2010 · Study method: LSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Strong**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Ochfeld2010**

▶ **Study context, limitations, review and source passages**

In chronic stroke, ischemia involving Broca area is not reliably associated with Broca or global aphasia.

@Ochfeld2010 · Study method: LSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
No Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Ochfeld2010**

▶ **Study context, limitations, review and source passages**

Longitudinal data shows that ischemia in Broca area predicts Broca or global aphasia acutely, but this association disappears by 6 months post-stroke.

@Ochfeld2010 · Study method: LSM · longitudinal

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Mixed Results	**Strong**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Ochfeld2010**

▶ **Study context, limitations, review and source passages**

Damage to the left inferior frontal gyrus pars opercularis (Brodmann area 44) is a strong predictor of Broca's aphasia (predictive relevance Z = 2.45).

@Yourganov2015Predicting · Study method: MLPA · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Strong**	**High**

Entry status: approved · Review provenance: legacy approval · Reviewer: auto-reviewer

▶ **Full citation for @Yourganov2015Predicting**

▶ **Study context, limitations, review and source passages**

Verbal quantity 1 finding

Damage to the left inferior frontal gyrus pars opercularis is associated with reduced verbal quantity in connected speech.

@Alyahya2020 · Study method: VBCM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Alyahya2020**

▶ **Study context, limitations, review and source passages**

Syntactic Processing Deficits in Aphasia 2 findings

Damage to the left inferior frontal gyrus pars opercularis (BA44) was not found to significantly affect sentence comprehension on the CYCLE-R.

@Dronkers2004 · Study method: VLSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
No Association	**Strong**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Dronkers2004**

▶ **Study context, limitations, review and source passages**

Damage to Broca's area (including pars opercularis and pars triangularis) is not causally associated with agrammatic sentence comprehension patterns in chronic patients.

@Rogalsky2018 · Study method: VLSM, behavioral_only · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
No Association	**Strong**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Rogalsky2018**

▶ **Study context, limitations, review and source passages**

Speech repetition 1 finding

Structural damage to Broca's area is associated with impaired speech repetition.

@Fridriksson2010 · Study method: VLSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Fridriksson2010**

▶ **Study context, limitations, review and source passages**

Anomic aphasia 2 findings

Increased activation in the left pars opercularis following treatment is associated with improved naming performance.

@Fridriksson2010Anomia · Study method: fMRI_activation · longitudinal

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Fridriksson2010Anomia**

▶ **Study context, limitations, review and source passages**

Restored blood flow to left BA 44/45 (Broca's area) independently contributes to the prediction of early improvement in picture naming after stroke.

@Hillis2006 · Study method: LSM · longitudinal

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Hillis2006**

▶ **Study context, limitations, review and source passages**

Apraxia of speech 1 finding

Dysfunction (infarct or hypoperfusion) of the left posterior inferior frontal gyrus (Broca's area) is strongly associated with apraxia of speech in acute stroke.

@Hillis2004 · Study method: LSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Strong**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Hillis2004**

▶ **Study context, limitations, review and source passages**

Phonological dyslexia 2 findings

Damage to the left inferior frontal gyrus pars opercularis is associated with phonological dyslexia.

@Rapcsak2009 · Study method: LSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Rapcsak2009**

▶ **Study context, limitations, review and source passages**

Damage to the left inferior frontal gyrus pars opercularis is associated with a larger lexicality effect in reading.

@Rapcsak2009 · Study method: LSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Rapcsak2009**

▶ **Study context, limitations, review and source passages**

Form-to-articulation processing (dorsal stream) 1 finding

Damage to the left inferior frontal gyrus pars opercularis loads strongly on the form-to-articulation (dorsal-stream) processing component in chronic post-stroke aphasia, identifying it as part of the dorsal speech-processing stream that supports motor-phonological aspects of speech production.

@Fridriksson2018 · Study method: VLSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Strong**	**High**

Entry status: approved · Review provenance: legacy approval · Reviewer: auto-reviewer

▶ **Full citation for @Fridriksson2018**

▶ **Study context, limitations, review and source passages**

Left Parietal Association Cortex (left)

10.2% of the lesion · 58% of the region destroyed · 1 distinct finding

Atlas match includes a documented judgement call | Limited citation base: 1 citation key

Semantics (semantic processing) 1 finding

Damage to the left parietal association cortex contributes significantly to impaired auditory comprehension.

@Bates2003 · Study method: VLSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Bates2003**

▶ **Study context, limitations, review and source passages**

Left Uncinate Fasciculus (left)

2.6% of the lesion · 57% of the region destroyed · 9 distinct findings

Semantics (semantic processing) 4 findings

No significant correlation was found between the integrity of the left uncinate fasciculus and naming deficits in this cohort.

@Breier2008 · Study method: DTI · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
No Association	**Weak**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Breier2008**

▶ **Study context, limitations, review and source passages**

The left uncinate fasciculus is implicated in lexical retrieval, semantic associations, and the naming of actions.

@Catani2008 · Study method: narrative_review · narrative-review

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Weak**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Catani2008**

▶ **Study context, limitations, review and source passages**

The uncinate fasciculus is involved in semantic processing and auditory working memory/sound recognition.

@Dick2012 · Study method: narrative_review · narrative-review

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Weak**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Dick2012**

▶ **Study context, limitations, review and source passages**

Damage to the left uncinate fasciculus is associated with deficits in object semantic processing.

@Han2013 · Study method: LSM, DTI · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Strong**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Han2013**

▶ **Study context, limitations, review and source passages**

Fluency (speech fluency) 2 findings

Damage to the left uncinate fasciculus predicts impaired speech fluency, particularly when controlling for motor speech agility.

@Fridriksson2013 · Study method: VBCM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Fridriksson2013**

▶ **Study context, limitations, review and source passages**

Damage to the left uncinate fasciculus is associated with deficits in both semantic and phonological verbal fluency.

@Li2017 · Study method: LSM, DTI · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Li2017**

▶ **Study context, limitations, review and source passages**

Form-to-meaning processing (ventral stream) 2 findings

Damage to the left uncinate fasciculus is a strong predictor of auditory comprehension impairment in patients with Broca's or global aphasia.

@Fridriksson2013 · Study method: VBCM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Fridriksson2013**

▶ **Study context, limitations, review and source passages**

Damage to the left uncinate fasciculus loads strongly on the form-to-meaning (ventral-stream) component — over 85% of uncinate-fasciculus voxels carry positive loadings — identifying it as the primary white-matter conduit of the ventral speech-processing stream.

@Fridriksson2018 · Study method: VLSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Strong**	**High**

Entry status: approved · Review provenance: legacy approval · Reviewer: auto-reviewer

▶ **Full citation for @Fridriksson2018**

▶ **Study context, limitations, review and source passages**

Global aphasia 1 finding

Damage to the left uncinate fasciculus contributes to global aphasia classification (predictive relevance Z = 1.71).

@Yourganov2015Predicting · Study method: MLPA · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: approved · Review provenance: legacy approval · Reviewer: auto-reviewer

▶ **Full citation for @Yourganov2015Predicting**

▶ **Study context, limitations, review and source passages**

Left Inferior Fronto-Occipital Fasciculus (left)

0.1% of the lesion · 56% of the region destroyed · 5 distinct findings

Also recorded in the knowledge base as: Left Inferior Frontal-Occipital Fasciculus

Semantics (semantic processing) 4 findings

Integrity of the left inferior frontal-occipital fasciculus is significantly related to language comprehension in chronic aphasia.

@Ivanova2016 · Study method: DTI · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Ivanova2016**

▶ **Study context, limitations, review and source passages**

The left inferior fronto-occipital fasciculus may be involved in reading and writing.

@Catani2008 · Study method: narrative_review · narrative-review

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Weak**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Catani2008**

▶ **Study context, limitations, review and source passages**

The inferior fronto-occipital fasciculus is involved in semantic processing, as evidenced by semantic paraphasias during electrostimulation.

@Dick2012 · Study method: narrative_review · narrative-review

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Weak**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Dick2012**

▶ **Study context, limitations, review and source passages**

Damage to the left inferior fronto-occipital fasciculus is associated with deficits in object semantic processing across visual and auditory modalities.

@Han2013 · Study method: LSM, DTI · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Strong**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Han2013**

▶ **Study context, limitations, review and source passages**

Fluency (speech fluency) 1 finding

Damage to the left inferior fronto-occipital fasciculus is associated with deficits in both semantic and phonological verbal fluency.

@Li2017 · Study method: LSM, DTI · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Li2017**

▶ **Study context, limitations, review and source passages**

Left IFG Pars Triangularis (BA 45) (left)

1.3% of the lesion · 52% of the region destroyed · 10 distinct findings

Naming 1 finding

Structural disconnection of left BA 45 (pars triangularis), independent of BA 45 cortical necrosis, is associated with poorer naming performance on the Philadelphia Naming Test in chronic aphasia after left-hemisphere stroke.

@Bonilha2014 · Study method: disconnectome · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: approved · Review provenance: human · Reviewer: michele

▶ **Full citation for @Bonilha2014**

▶ **Study context, limitations, review and source passages**

Fluency (speech fluency) 2 findings

Structural disconnection of left BA 45 (pars triangularis), independent of BA 45 cortical necrosis, is independently associated with reduced speech fluency in chronic aphasia after left-hemisphere stroke; BA 45 necrosis alone is not an independent predictor of fluency.

@Bonilha2014 · Study method: disconnectome · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: approved · Review provenance: human · Reviewer: michele

▶ **Full citation for @Bonilha2014**

▶ **Study context, limitations, review and source passages**

Damage to the left IFG pars triangularis (BA45) does not significantly predict long-term speech production scores.

@GajardoVidal2021 · Study method: RLSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
No Association	**Strong**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @GajardoVidal2021**

▶ **Study context, limitations, review and source passages**

Verbal quality 1 finding

Damage to the left inferior frontal gyrus pars triangularis is associated with reduced verbal quality in connected speech.

@Alyahya2020 · Study method: VBCM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Alyahya2020**

▶ **Study context, limitations, review and source passages**

Syntactic Processing Deficits in Aphasia 2 findings

Damage to the left inferior frontal gyrus pars triangularis (BA47) is associated with deficits in the comprehension of complex sentences, particularly those involving relative clauses.

@Dronkers2004 · Study method: VLSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Strong**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Dronkers2004**

▶ **Study context, limitations, review and source passages**

Damage to the left inferior frontal gyrus pars triangularis is significantly associated with agrammatism, independent of speech rate.

@Matchin2020 · Study method: LSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Strong**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Matchin2020**

▶ **Study context, limitations, review and source passages**

Aphasia severity metric (instrument-agnostic) 2 findings

In the acute phase post-stroke, better initial language performance is positively correlated with higher activation in the left inferior frontal gyrus.

@Saur2006 · Study method: fMRI_activation · longitudinal

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Saur2006**

▶ **Study context, limitations, review and source passages**

Higher activation in the left inferior frontal gyrus pars triangularis is associated with better language comprehension across acute, subacute, and chronic phases.

@Stockert2020 · Study method: fMRI_activation · longitudinal

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Stockert2020**

▶ **Study context, limitations, review and source passages**

Semantics (semantic processing) 1 finding

Damage to the left lateral prefrontal cortex, including the inferior frontal gyrus (BA 45), is associated with semantic naming errors, but this effect is mediated by conceptualization processes.

@Schwartz2009 · Study method: VLSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Schwartz2009**

▶ **Study context, limitations, review and source passages**

Global aphasia 1 finding

Damage to the left inferior frontal gyrus pars triangularis (BA 45) contributes to global aphasia classification (predictive relevance Z = 2.18).

@Yourganov2015Predicting · Study method: MLPA · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Strong**	**High**

Entry status: approved · Review provenance: legacy approval · Reviewer: auto-reviewer

▶ **Full citation for @Yourganov2015Predicting**

▶ **Study context, limitations, review and source passages**

Left Supramarginal Gyrus and Premotor Cortex (supra-Sylvian) (left)

11.2% of the lesion · 50% of the region destroyed · 3 distinct findings

Atlas match includes a documented judgement call

Speech production phonological 1 finding

Lesions in the left supramarginal gyrus and adjacent supra-Sylvian cortex (extending anteriorly into inferior postcentral, precentral, and premotor cortex) are associated with phonological/articulatory speech production deficits in chronic post-stroke aphasia, as captured by a Speech Production factor loading on word repetition, nonword repetition, and phonological error rate in picture naming.

@Mirman2015 · Study method: VLSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Strong**	**High**

Entry status: approved · Review provenance: legacy approval · Reviewer: claude-cowork

▶ **Full citation for @Mirman2015**

▶ **Study context, limitations, review and source passages**

Phonological production 1 finding

Damage to the left supramarginal gyrus is associated with impaired phonematic structure in spontaneous speech and impaired speech repetition.

@Henseler2014 · Study method: VLSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: approved · Review provenance: model · Reviewer: model:gemma4:26b · Review tier: A

▶ **Full citation for @Henseler2014**

▶ **Study context, limitations, review and source passages**

Anomic aphasia 1 finding

Increased activation in the left inferior parietal lobule following treatment is associated with improved naming performance.

@Fridriksson2010Anomia · Study method: fMRI_activation · longitudinal

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Fridriksson2010Anomia**

▶ **Study context, limitations, review and source passages**

Left Inferior Longitudinal Fasciculus (left)

0.1% of the lesion · 47% of the region destroyed · 5 distinct findings

Semantics (semantic processing) 4 findings

The left inferior longitudinal fasciculus is likely involved in visual object recognition and linking object representations to lexical labels.

@Catani2008 · Study method: narrative_review · narrative-review

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Weak**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Catani2008**

▶ **Study context, limitations, review and source passages**

Impaired auditory word-picture matching is associated with damage to the left inferior longitudinal fasciculus.

@Fridriksson2013 · Study method: VBCM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Fridriksson2013**

▶ **Study context, limitations, review and source passages**

Integrity of the left inferior longitudinal fasciculus is significantly related to language comprehension in chronic aphasia.

@Ivanova2016 · Study method: DTI · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Ivanova2016**

▶ **Study context, limitations, review and source passages**

Tissue concentration in the left inferior longitudinal fasciculus uniquely covaries with the semantic PCA factor — connecting posterior occipito-temporal cortex with the temporal pole, consistent with the ventral language pathway.

@Alyahya2018NounVerb · Study method: VBCM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Weak**	**High**

Entry status: approved · Review provenance: human · Reviewer: michele

▶ **Full citation for @Alyahya2018NounVerb**

▶ **Study context, limitations, review and source passages**

Phonological production 1 finding

Tissue concentration in the left inferior longitudinal fasciculus uniquely covaries with the phonological production PCA factor in chronic post-stroke aphasia.

@Alyahya2018NounVerb · Study method: VBCM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: approved · Review provenance: legacy approval · Reviewer: auto-reviewer

▶ **Full citation for @Alyahya2018NounVerb**

▶ **Study context, limitations, review and source passages**

Left Postcentral Gyrus (left)

4.1% of the lesion · 28% of the region destroyed · 1 distinct finding

Limited citation base: 1 citation key

Apraxia of speech 1 finding

In a subset of patients with apraxia of speech who lack Broca's area dysfunction, dysfunction of the left postcentral gyrus (sensory cortex) is associated with the deficit.

@Hillis2004 · Study method: LSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Hillis2004**

▶ **Study context, limitations, review and source passages**

White-matter disconnection: what the evidence says

No disconnectome map for this lesion mask (run the BCBToolkit or DeepDisco step, or choose another map in the report settings).

This participant’s measures: what the evidence says

Findings about measures such as lesion volume or aphasia severity report associations in study groups (for example, larger lesions with more severe aphasia), not predictions for one person.

Aphasia severity	WAB-AQ 51.5 (moderate, Kertesz bands), measured 3839 days after the stroke

Lower values go with worse language or recovery. (14 distinct findings: 14 association)

▼ **Show 14 distinct findings**

Left perisylvian network 2 findings

Damage to left perisylvian regions — particularly the rolandic operculum, superior temporal gyrus, and inferior temporal gyrus — accounts for the largest shared variance across all four WAB-R subtests in chronic post-stroke aphasia, with the rolandic operculum implicated across seven out of eleven neuroimaging modality-specific PLSR models.

@Kristinsson2025 · Study method: MLPA, DTI, fMRI_FC · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: approved · Review provenance: human · Reviewer: human:michele

▶ **Full citation for @Kristinsson2025**

▶ **Study context, limitations, review and source passages**

After controlling for shared perisylvian lesion variance, right putamen integrity uniquely and independently predicts naming and repetition performance, while left superior temporal lobe and right precuneus integrity uniquely predict auditory comprehension.

@Kristinsson2025 · Study method: MLPA, DTI · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Weak**	**Medium**

Entry status: approved · Review provenance: human · Reviewer: human:michele

▶ **Full citation for @Kristinsson2025**

▶ **Study context, limitations, review and source passages**

Prognosis 5 findings

Lower initial severity of aphasia is associated with a higher degree of recovery.

@Laska2001 · Study method: behavioral_only · longitudinal

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: approved · Review provenance: model · Reviewer: model:gemma4:26b · Review tier: A

▶ **Full citation for @Laska2001**

▶ **Study context, limitations, review and source passages**

Higher initial aphasia severity (higher WAB-AQ) predicts a greater decline in language abilities over time in chronic aphasia.

@Basilakos2019 · Study method: behavioral_only · longitudinal

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Basilakos2019**

▶ **Study context, limitations, review and source passages**

Initial aphasia severity measured by the Aphasia Rapid Test at day 7 predicts 3-month functional outcome.

@Benghanem2019 · Study method: behavioral_only · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Strong**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Benghanem2019**

▶ **Study context, limitations, review and source passages**

Initial severity of the language deficit is a significant predictor of subsequent language recovery.

@Hartwigsen2019 · Study method: narrative_review · narrative-review

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Strong**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Hartwigsen2019**

▶ **Study context, limitations, review and source passages**

A combination of baseline lesion site and time post-stroke accurately predicts future speech production scores.

@Hope2013 · Study method: MLPA · longitudinal

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Strong**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Hope2013**

▶ **Study context, limitations, review and source passages**

Rehabilitation response 2 findings

Baseline aphasia severity (WAB-AQ) independently predicts responsiveness to language rehabilitation in chronic stroke.

@Billot2022 · Study method: MLPA · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Billot2022**

▶ **Study context, limitations, review and source passages**

A multimodal combination of aphasia severity, demographics, structural integrity, and resting-state connectivity predicts treatment response significantly better than any single feature set.

@Billot2022 · Study method: MLPA · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Strong**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Billot2022**

▶ **Study context, limitations, review and source passages**

Anomic aphasia 1 finding

Lower baseline aphasia severity (higher WAB-AQ) is associated with greater improvement in naming following speech therapy.

@Bonilha2016 · Study method: behavioral_only · longitudinal

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Bonilha2016**

▶ **Study context, limitations, review and source passages**

Alexia 1 finding

Lower aphasia severity (higher WAB-AQ scores) significantly predicts better oral reading performance across total, lexical, and sublexical reading.

@Brookshire2014 · Study method: behavioral_only · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Brookshire2014**

▶ **Study context, limitations, review and source passages**

Transcranial Direct Current Stimulation (tDCS) during aphasia treatment 1 finding

Baseline aphasia severity, measured by the WAB-R Aphasia Quotient, is positively correlated with the improvement in naming following aphasia treatment.

@Fridriksson2018JAMA · Study method: behavioral_only · RCT

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Fridriksson2018JAMA**

▶ **Study context, limitations, review and source passages**

Phonological Component Analysis (PCA) 1 finding

Less severe initial naming impairment (higher BNT scores) is associated with a better response to Phonological Components Analysis (PCA) treatment.

@Leonard2008 · Study method: behavioral_only · case-series

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Leonard2008**

▶ **Study context, limitations, review and source passages**

Phonological dyslexia 1 finding

General phonological ability (PC score) is a strong predictor of reading accuracy in patients with perisylvian lesions.

@Rapcsak2009 · Study method: behavioral_only · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Strong**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Rapcsak2009**

▶ **Study context, limitations, review and source passages**

□ 10 of these findings concern severity measured early after the stroke (or recovery from it); this score was measured 3839 days after the stroke, so they apply only loosely.

Age at stroke 54 years

Higher values go with worse language or recovery. (7 distinct findings: 3 association, 2 inverse association, 2 no association)

▼ **Show 7 distinct findings**

Aphasia severity metric (instrument-agnostic) 3 findings

Older age at stroke onset is associated with long-term decline in aphasia severity in chronic stroke survivors.

@Johnson2019 · Study method: behavioral_only · longitudinal

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Inverse Association	**Moderate**	**High**

Entry status: approved · Review provenance: model · Reviewer: model:gemma4:26b · Review tier: A

▶ **Full citation for @Johnson2019**

▶ **Study context, limitations, review and source passages**

Working-age participants (<=65 years) achieve greater overall-language gains with high-intensity SLT (>9 h/week), whereas older participants (>65 years) achieve greatest gains with low-intensity SLT (<=2 h/week).

@Brady2022 · Study method: meta-analysis

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Brady2022**

▶ **Study context, limitations, review and source passages**

Older age is associated with worse language recovery (lower WAB-R AQ) at 6 months post-stroke.

@Forkel2014 · Study method: tractography · longitudinal

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Inverse Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Forkel2014**

▶ **Study context, limitations, review and source passages**

Anomic aphasia 1 finding

Age at stroke onset does not predict the direction of language recovery or decline in chronic aphasia.

@Hope2017 · Study method: behavioral_only · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
No Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Hope2017**

▶ **Study context, limitations, review and source passages**

Prognosis 2 findings

Younger age is associated with a greater degree of language recovery following acute stroke.

@Laska2001 · Study method: behavioral_only · longitudinal

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Laska2001**

▶ **Study context, limitations, review and source passages**

Age is not a clinically relevant independent predictor of aphasia recovery outcome.

@Pedersen1995 · Study method: behavioral_only · longitudinal

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
No Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Pedersen1995**

▶ **Study context, limitations, review and source passages**

Aphasia severity 1 finding

Older age at the time of stroke is associated with higher rates of aphasia.

@Wallentin2018 · Study method: behavioral_only · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Strong**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Wallentin2018**

▶ **Study context, limitations, review and source passages**

Time since stroke 3839 days

Direction: see each finding. (4 distinct findings: 3 association, 1 no association)

▼ **Show 4 distinct findings**

Aphasia severity metric (instrument-agnostic) 1 finding

Participants within 3 months of onset achieve greatest overall-language gains with low-intensity SLT (<=2 h/week), whereas those >3 months post-onset achieve greatest gains with moderate-intensity SLT (3-4 h/week).

@Brady2022 · Study method: meta-analysis

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Brady2022**

▶ **Study context, limitations, review and source passages**

Fluency (speech fluency) 1 finding

Time post-stroke is the most significant single predictor of speech production recovery.

@Hope2013 · Study method: MLPA · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Strong**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Hope2013**

▶ **Study context, limitations, review and source passages**

Anomic aphasia 1 finding

Time post-stroke at first assessment does not predict the direction of language recovery or decline in chronic aphasia.

@Hope2017 · Study method: behavioral_only · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
No Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Hope2017**

▶ **Study context, limitations, review and source passages**

Melodic Intonation Therapy (MIT) 1 finding

Earlier initiation of Melodic Intonation Therapy in the subacute phase predicts greater improvement in language repetition and verbal communication.

@vanderMeulen2014 · Study method: behavioral_only · longitudinal

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @vanderMeulen2014**

▶ **Study context, limitations, review and source passages**

Sex M

Direction: see each finding. (3 distinct findings: 2 association, 1 no association)

▼ **Show 3 distinct findings**

Aphasia severity metric (instrument-agnostic) 1 finding

Males achieve greatest overall-language and auditory comprehension gains with high-intensity SLT (>9 h/week), while females achieve greatest gains with low-intensity SLT (<2 h/week).

@Brady2022 · Study method: meta-analysis

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Brady2022**

▶ **Study context, limitations, review and source passages**

Aphasia severity 2 findings

Women have significantly higher rates of aphasia following stroke than men.

@Wallentin2018 · Study method: meta-analysis

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Wallentin2018**

▶ **Study context, limitations, review and source passages**

When controlling for age and stroke severity, sex does not independently predict aphasia rates.

@Wallentin2018 · Study method: behavioral_only · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
No Association	**Strong**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Wallentin2018**

▶ **Study context, limitations, review and source passages**

Lesion volume 199 ml (this report's lesion mask)

Higher values go with worse language or recovery. (19 distinct findings: 13 association, 5 no association, 1 inverse association)

▼ **Show 19 distinct findings**

Aphasia severity metric (instrument-agnostic) 11 findings

Total lesion volume strongly correlates with single-word verb naming, noun naming, verb picture-to-word matching, and noun picture-to-word matching in chronic post-stroke aphasia (Pearson r = .56–.67, p<.0001 for all four measures).

@Alyahya2018NounVerb · Study method: VBCM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Strong**	**High**

Entry status: approved · Review provenance: human · Reviewer: human:michele

▶ **Full citation for @Alyahya2018NounVerb**

▶ **Study context, limitations, review and source passages**

Smaller total lesion volume is associated with greater long-term WAB-R AQ improvement in chronic aphasia patients retested at least one year after initial chronic-stage assessment; lesion volume was the only significant demographic or stroke-related predictor in the regression model.

@Harvey2022 · Study method: VLSM · longitudinal

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: approved · Review provenance: human · Reviewer: human:michele

▶ **Full citation for @Harvey2022**

▶ **Study context, limitations, review and source passages**

Greater satisfaction with life participation (ALA participation domain) is associated with better long-term aphasia recovery (WAB-R AQ gain), independent of lesion volume, in chronic aphasia.

@Harvey2022 · Study method: behavioral_only · longitudinal

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Weak**	**Medium**

Entry status: approved · Review provenance: human · Reviewer: human:michele

▶ **Full citation for @Harvey2022**

▶ **Study context, limitations, review and source passages**

Larger lesion volume is associated with more severe initial aphasia (lower WAB-AQ) in chronic stroke.

@Basilakos2019 · Study method: LSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Basilakos2019**

▶ **Study context, limitations, review and source passages**

Total lesion volume accounts for a small amount of variance in language and other behavioral deficits in acute stroke.

@Corbetta2015 · Study method: MLPA · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Strong**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Corbetta2015**

▶ **Study context, limitations, review and source passages**

Larger left-hemisphere lesion volume is associated with worse language recovery (lower WAB-R AQ) at 6 months post-stroke.

@Forkel2014 · Study method: tractography · longitudinal

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Inverse Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Forkel2014**

▶ **Study context, limitations, review and source passages**

There is a strong coupling between impairment and lesion size, as larger lesions are more likely to affect critical anatomical structures.

@Karnath2018 · Study method: narrative_review · narrative-review

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Karnath2018**

▶ **Study context, limitations, review and source passages**

Total lesion volume is significantly associated with overall aphasia severity and specific language functions in chronic stroke.

@Kristinsson2021 · Study method: behavioral_only · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Strong**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Kristinsson2021**

▶ **Study context, limitations, review and source passages**

Overall lesion size is a significant predictor of general aphasia severity (WAB-AQ) and object naming (PNT) accuracy.

@Thye2018 · Study method: VLSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Strong**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Thye2018**

▶ **Study context, limitations, review and source passages**

Lesion size is a stronger predictor of broad language deficits (WAB-AQ, PNT) than of specific functional language systems (Speech Production, Speech Recognition).

@Thye2018 · Study method: VLSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Strong**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Thye2018**

▶ **Study context, limitations, review and source passages**

Overall lesion size is a significant predictor of WAB scores, including fluency, auditory comprehension, repetition, naming, and the aphasia quotient.

@Yourganov2016 · Study method: behavioral_only · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Strong**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Yourganov2016**

▶ **Study context, limitations, review and source passages**

Fluency (speech fluency) 2 findings

Total lesion volume strongly predicts the fluency PCA factor in chronic post-stroke aphasia (r=.59, p<.0001), and adding lesion volume as a covariate eliminates all significant lesion-symptom clusters for this factor.

@Alyahya2018NounVerb · Study method: VBCM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Strong**	**High**

Entry status: approved · Review provenance: human · Reviewer: human:michele

▶ **Full citation for @Alyahya2018NounVerb**

▶ **Study context, limitations, review and source passages**

Total lesion volume is a significant predictor of speech production impairment.

@Hope2013 · Study method: MLPA · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Hope2013**

▶ **Study context, limitations, review and source passages**

Phonological production 1 finding

Total lesion volume does NOT significantly correlate with the phonological production PCA factor in chronic post-stroke aphasia (r=.22, p=.12); the lesion-symptom cluster for this factor is robust to lesion-volume control.

@Alyahya2018NounVerb · Study method: VBCM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
No Association	**Moderate**	**High**

Entry status: approved · Review provenance: human · Reviewer: human:michele

▶ **Full citation for @Alyahya2018NounVerb**

▶ **Study context, limitations, review and source passages**

Prognosis 3 findings

Lesion volume does not significantly predict whether a patient recovers completely from aphasia.

@Laska2001 · Study method: LSM · longitudinal

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
No Association	**Moderate**	**High**

Entry status: approved · Review provenance: model · Reviewer: model:gemma4:26b · Review tier: A

▶ **Full citation for @Laska2001**

▶ **Study context, limitations, review and source passages**

Total infarct volume is a univariate predictor of 3-month functional outcome.

@Benghanem2019 · Study method: LSM · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Benghanem2019**

▶ **Study context, limitations, review and source passages**

Lesion size does not significantly correlate with initial language severity, 90-day language functioning, or total language recovery.

@Lazar2008 · Study method: behavioral_only · longitudinal

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
No Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Lazar2008**

▶ **Study context, limitations, review and source passages**

Anomic aphasia 2 findings

Total necrotic lesion volume is not associated with treatment-induced improvements in naming.

@Bonilha2016 · Study method: LSM · longitudinal

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
No Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Bonilha2016**

▶ **Study context, limitations, review and source passages**

Lesion volume does not predict the direction of language recovery or decline in chronic aphasia.

@Hope2017 · Study method: behavioral_only · cross-sectional

RELATIONSHIP	EVIDENCE STRENGTH	EXTRACTION CONFIDENCE
No Association	**Moderate**	**High**

Entry status: draft · Review provenance: no named reviewer

▶ **Full citation for @Hope2017**

▶ **Study context, limitations, review and source passages**

Therapy recommendations (5 matched)

These are therapies with evidence targeting impairments represented above. They are research associations, not prescriptions; recovery-stage mismatches remain marked for clinical review.

Therapy	Addresses impairment(s)	Active ingredients (RTSS)	Brief description	Citations
Constraint-Induced Aphasia Therapy (CIAT / CILT)	**Broca's aphasia** (behavioural); Anomic aphasia, Broca's aphasia, Fluency (speech fluency), Form-to-meaning processing (ventral stream), Global aphasia (lesion)	Verbal Constraint; Communicative Shaping; Massed Practice	An intensive, massed-practice therapy in which compensatory strategies (gesture, drawing, writing) are prohibited and th…	@Maher2006, @Pulvermuller2001
Melodic Intonation Therapy (MIT)	**Broca's aphasia** (behavioural); Broca's aphasia, Fluency (speech fluency), Global aphasia, Phonological production (lesion)	Melodic Intoning; Rhythmic Left-Hand Tapping; Communicative Shaping; Massed Practice	A therapy that uses melodic intoning (singing on two pitches) and rhythmic left-hand tapping to elicit propositional spe…	@Schlaug2008, @Stahl2013, @vanderMeulen2012, @vanderMeulen2014
Melodic Intonation Therapy	**Broca's aphasia** (behavioural); Fluency (speech fluency), Phonological production (lesion)	—	A language production treatment for severe nonfluent aphasia involving repetitive singing of short sentences combined wi…	@Schlaug2008, @Stahl2013, @vanderMeulen2012, @vanderMeulen2014
Treatment of Underlying Forms (TUF)	**Broca's aphasia** (behavioural); Broca's aphasia, Syntactic Processing Deficits in Aphasia (lesion)	Verb Argument Structure Training	A linguistically-motivated, impairment-based intervention for agrammatic aphasia that uses meta-linguistic steps focused…	@Barbieri2023, @Thompson2003
Transcranial Direct Current Stimulation (tDCS) during aphasia treatment	**Broca's aphasia** (behavioural); Anomic aphasia, Broca's aphasia, Fluency (speech fluency) (lesion)	Anodal tDCS (Left Temporoparietal)	Anodal tDCS applied to the intact left temporoparietal region for the first 20 min of each behavioral aphasia treatment …	@Fridriksson2018BDNF, @Fridriksson2018JAMA

procedure dose parameter technology

How to read the evidence

Display signal is CALMaR's project-specific ordering heuristic, not a formal evidence grade or participant-level prediction. Its basis is explained beside the outcome summaries.

Relationship is what a study reported between the region or measure and the outcome: association, inverse association, no association or mixed results. **Evidence strength** is the extractor's reading of the finding. **Extraction confidence** is confidence that the paper was transcribed and interpreted correctly. These are distinct.

Finding counts are descriptive. They do not measure certainty, and findings do not necessarily represent independent studies or independent cohorts. Group associations do not predict this participant's outcome.

An "atlas match includes a documented judgement call" note links a knowledge-base region to atlas units using a curated decision recorded in `aphasia-kb/atlas_mapping/README.md`.

WAB decomposition: flu = fluency, comp = comprehension, rep = repetition. + preserved, − impaired (WAB type taxonomy).
Stage badge: Acute <2 wk · Subacute 2 wk–6 mo · Chronic >6 mo. A therapy-stage warning identifies evidence from a different recovery stage; open it for detail. Rules and their sources: `aphasia-kb/aphasia_rules.py`.